A Fixed CM Packet on McMullen's Curve and Its Principal Models

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Abstract

We determine the intersection of a fixed marked CM norm packet with McMullen's normalized curve. Its arithmetic census has 3520 trace-labelled integral matrix classes and 368 quotient points. Under explicitly stated historical execution assumptions, exactly three quotient points occur on the curve. We distinguish their nonprincipal trace polarizations from a common self-dual lattice modification, which gives principal models on a connected finite-cover RM family. The three principal models remain distinct after forgetting polarizations and markings. For the B/C models, the minimum unrestricted isogeny degree is 7, while the minimum scalar-polarized degree is 64. A unit twist preserves the principal curve image and gives six distinct principal Weil points. The Hodge consequences concern the specified fibers; identification with a differently normalized principal moduli locus remains open.

Companion references. Result numbers prefixed I, II, H and S refer respectively to the packet paper, quotient paper, supporting Hodge note and computational supplement. Each document has its own numbering and bibliography.

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1 Introduction

A rational CM Hodge structure does not determine an integral polarized abelian variety, and neither determines its location on a specified curve. This paper separates these three questions for one finite packet on McMullen's curve. We first determine its intersection with the normalized curve. We then construct principal models of the three surviving fibers and distinguish their integral maps and polarizations.

The packet is fixed by the extension M/L, the ordered embeddings and the norm generator below. All conjugacy counts use $\mathrm{SL}_2(\mathcal{O}_L)$. Replacing this group by $\mathrm{GL}_2(\mathcal{O}_L)$ or forgetting a field marking changes the classification problem. No assertion concerns every CM point of the curve or every Hecke packet of level 43.

1.1 The three questions and their answers

The finite packet has 3520 trace-labelled matrix classes and 368 quotient points. Its intersection with the normalized curve is exactly A, B, C, subject to the execution premise stated below. The original fibers have polarization type $(1, 7, 7, 21, 21, 21)$, not principal type. A common self-dual lattice gives isogenous principal models on a connected finite cover. For the B/C principal models the minimum unrestricted isogeny degree is 7, whereas the minimum scalar-polarized degree is 64. These two minima measure different requirements on a map.

The table in Theorem 1.2 presents all three geometric fibers together. The main statements distinguish the finite intersection, the Hodge consequences and the integral principal geometry. The supporting Hodge note treats powers and mixed products; none of those extensions is needed to prove the finite intersection.

1.2 Fixed data and execution premise

Write $z = \zeta_{42}$, $M = \mathbb{Q}(z) = \mathbb{Q}(\zeta_{21})$, $L = M^+$ and $t = z + z^{-1}$. Order the real embeddings by $1, 5, 11, 13, 17, 19$ and put

$$\pi = t^4 + t^3 - 4t^2 - 2t + 5, \qquad \alpha_0 = 1 + z + z^4, \qquad \alpha_k = z^k \alpha_0, \quad c_k = \alpha_k + \bar{\alpha}_k.$$

Then $N_{M/L}(\alpha_0) = \pi$ and $N_{L/\mathbb{Q}}(\pi) = 43$. The symbols A, B, C denote normalized vertex classes; A_X denotes the trace-polarized sixfold at X, and B_X its specified principal model. Let $F_t : \mathbb{H} \to \mathbb{H}^6$ be McMullen's modular embedding with the component normalization of Supplement, Section S.7. The normalized curve image is the image of F_t in $\mathrm{SL}_2(\mathcal{O}_L)\backslash\mathbb{H}^6$. Its quadrilateral monodromy is the integral, orientation-preserving group used below. The normalized generators are $X_t = h_t X h_t^{-1}$ with $h_t = \mathrm{diag}(t, 1)$. The fourth quadrilateral vertex is treated in Proposition 5.17; it is not a fourth quotient class.

Packet execution hypothesis. The packet intersection claims use the retained exact search executions identified in Supplement, Section S.13. For the historical commands and continuation cohort lacking complete end-of-execution guards, we assume that the archived mathematical sources and runtime were not transiently replaced during execution. The accepted source correspondence and final population join are those of the final packet adoption identified there. Later hashes and small control runs do not establish the missing historical guards or constitute a fresh execution of the full suite. This premise concerns the retained execution evidence; it is additional to the finite-domain and exact-arithmetic arguments in the proofs.

Theorem 1.1 (The fixed marked norm-π packet)**.** *Under the packet execution hypothesis above, the following holds. Fix the extension M/L, its L-action and ordered six real embeddings, and the generator π and elements $\alpha_k = \zeta_{42}^k \alpha_0$ for $0 \le k < 42$ specified above. The integral matrices of determinant π and trace $c_k = \alpha_k + \overline{\alpha}_k$ have exactly* 3520 *trace-labelled* $\mathrm{SL}_2(\mathcal{O}_L)$*-conjugacy classes. Their upper fixed tuples give exactly* 368 *distinct points of* $\mathrm{SL}_2(\mathcal{O}_L)\backslash\mathbb{H}^6$*. In the normalized trace-polarized McMullen curve image, precisely three of these points occur: A, B, C. There are* 365 *exclusions, consisting of thirteen maximal-multiplier points and all* 352 *proper-multiplier points. The complete* 112*-point $d = 7$ subpacket has three hits and* 109 *exclusions.*

Proof. The arithmetic assertion is Theorem 3.3. The geometric assertion is Theorem 3.6, using the retained exact certificates and execution-provenance assumptions specified in Supplement, Section S.13. □

Theorem 1.2. *Let z_X be the normalized vertex $X \in \{A, B, C\}$ and set $A_X = \mathbb{C}^6/(\mathcal{O}_L z_X + \mathcal{O}_L)$, with trace polarization $E_1((a,b),(a',b')) = \mathrm{Tr}_{L/\mathbb{Q}}(a'b - ab')$. Each has polarization type $(1,7,7,21,21,21)$ and a compatible $K = \mathbb{Q}(\sqrt{-3})$ action of signature $(3,3)$ and Hermitian discriminant -1. Their actual geometric CM types and Hodge algebras are*

X	Φ_X	$\mathrm{Hdg}^*(A_X)$
A	$\{1,5,17,19,29,31\}$	$\langle \mathit{divisors} \rangle$
B	$\{1,5,23,25,29,31\}$	$\langle \mathit{divisors}, W_K(A_B) \rangle$
C	$\{1,5,11,13,17,19\}$	$\langle \mathit{divisors}, W_K(A_C) \rangle$

*The A type is induced and the B/C types are primitive. Consequently the Hodge conjecture holds for all three fibers in every codimension. Their common self-dual lattice modification supplies principal models B_A, B_B, B_C on one connected finite-cover RM family, with degree-*453789 *isogenies preserving the rational CM Hodge structures, the indicated K signatures and discriminants.*

Proof locator for Theorem 1.2. The trace polarization is computed in Proposition 4.1. The B calculation is in Sections 5.4–5.6; Theorem 5.14 and Proposition 5.15 give A and C. Proposition 6.1 and Theorem 6.4 supply the common principal family.

Theorem 1.3 (Principal models and their integral distinctions)**.** *The specified principal models B_A, B_B, B_C are pairwise nonisomorphic even as unpolarized complex abelian varieties. For maps $B_B \to B_C$, the least isogeny degree is* 7*. Among these maps pulling back the given target principal polarization to a positive integer multiple of the source polarization, the least multiplier is* 2*, with exactly six degree-*64 *maps sharing one kernel. Under the normalization of Proposition 7.8, their integral Hom module is the target-R ideal $\mathfrak{a}$ of class order two.*

On each of the two underlying varieties B_B, B_C, exactly two isomorphism classes of principal polarizations occur, represented by the given polarization and its $4 - t^2$ twist. These give four distinct polarized isomorphism classes after forgetting all CM and RM markings. The same twist extends over the principal family; on the full coarse principal RM component it is the nontrivial involution $w_i \mapsto w_i/\sigma_i(4-t^2)$. It preserves the entire principal curve image. On each of B_A, B_B, B_C with its prescribed RM action there are exactly two compatible principal-polarization classes; the resulting six Weil points all lie on this same image and remain distinct after forgetting all CM and RM markings. For each of the four B/C polarization pairs, the least scalar multiplier remains 2*, with six degree-*64 *maps in the direction $B_B \to B_C$; their two possible kernels are complementary order-*64 *subgroups of $B_B[2]$.*

Proof. See Theorem 6.4 and Propositions 7.6, 7.5, 7.7, 7.8, 7.12 and 8.1, together with Propositions 8.2, 8.3, 8.4, 8.5 and Corollary 8.7. Arbitrary fine-level lifts are not part of the involution assertion. □

Remark 1.4 (Principal-moduli normalization). The original lattice $\mathcal{O}_L^2$ with its trace form is not a principally polarized model. The self-dual modification changes the lattice through an isogeny; passage to a finite cover alone would not do so. The resulting principal family lies in the component determined by that self-dual module. Its identification with Mostaed's named principally polarized locus remains open. The Hodge conclusions at the three special fibers do not require that further identification, and no global K-action on the family is asserted.

1.3 Proof strategy and relation to prior work

The arithmetic classification includes every module overorder and orientation, then identifies the two characteristic roots at the level of quotient points. For geometry, two simultaneous target regions reduce curve membership to finite exact searches. Both regions are needed: separate choices of a reflected chamber in different coordinates are not allowed. The positive vertex realizations and exhaustive empty searches prove the intersection theorem.

The principal construction instead uses the discriminant module of the trace lattice. A common nonfree maximal isotropic subgroup is stable under the three specialized quadratic actions. Its stabilizer gives the finite cover on which the lattice modification descends. Integral lattice and unit calculations then distinguish the models, maps and twists.

The Hodge consequences use divisor classes and the split-Weil input of Schoen and Markman [13, 4]. OpenAI's general CM manuscript [11], if valid, implies the rational algebraicity conclusions; it is not a proof input here. Goodson's factor comparison and Saller's related discussion [2, 12] also precede any claim of novelty for the rational CM types. The contribution examined here is the fixed packet and its integral geometry, not new abstract isogeny classes. Precise historical priority remains a separate literature question. The reported Aoki route in Saller is not used as a substitute proof.

The trace-dual and unit-norm description of polarizations is classical. For comparison, Li–Mantovan–Pries prove uniqueness, when a principal polarization exists, for simple maximal-order CM types under specified unit-sign hypotheses [3, Prop. 4.5]. Their same-field example $y^7 = x^3 - 1$ is a known principally polarized Jacobian [3, Prop. 6.11 and Lemma 6.12]. Our B/C principal models have a proper endomorphism order. The results here concern their exact integral classes, maps and realization on the specified curve; neither the general polarization method nor same-field principal existence is claimed as new.

The constructed principal family is not identified with Mostaed's named principal normalization [10]. It has a different integral lattice. Corrections relevant to that distinction are recorded in the supplement; they are secondary to the positive results of this paper.

2 Arithmetic and Hilbert-modular setup

Let

$$M = \mathbb{Q}(\zeta_{42}), \qquad L = M^+ = \mathbb{Q}(\zeta_{42} + \zeta_{42}^{-1}),$$

so $[L : \mathbb{Q}] = 6$ and $[M : L] = 2$. Put

$$t = \zeta_{42} + \zeta_{42}^{-1} = 2\cos(\pi/21).$$

Then t has minimal polynomial

$$f(T) = T^6 + T^5 - 6T^4 - 6T^3 + 8T^2 + 8T + 1. \tag{2.1}$$

We use the six real embeddings indexed by

$$k \in \{1, 5, 11, 13, 17, 19\}, \qquad \sigma_k(t) = 2\cos(k\pi/21).$$

For a prime ideal $\mathfrak{l} \subset \mathcal{O}_L$ of absolute norm ℓ, the Hilbert-modular Hecke correspondence is represented by matrices whose determinant ideal is $\mathfrak{l}$. When $\mathfrak{l}$ is narrowly principal and a totally positive generator π is fixed, standard representatives have determinant π. This is also the determinant convention used explicitly in Mostaed's Section 5.3: if $\alpha = \delta\gamma_j$, then $\det_L(\alpha) = \pi$ [10, Sec. 5.3].

3 The complete packet: arithmetic and geometry

The proof separates the finite arithmetic domain from curve membership. The supplementary order calculations feed into the following full census; its proof includes noninvertible modules over the originally generated order. Historical partial exclusion totals are kept in the supplement and play no role as substitutes for the final two-target cover.

3.1 The complete marked arithmetic census

Throughout this subsection the extension M/L, its L-action and the ordered six real embeddings are fixed. Write $z = \zeta_{42}$ and

$$\alpha_k = z^k(1 + z + z^4) = A_k + B_k z, \qquad c_k = 2A_k + tB_k, \qquad 0 \le k < 42.$$

We count integral matrices of trace c_k and determinant π, up to $\mathrm{SL}_2(\mathcal{O}_L)$ conjugacy, and their upper fixed tuples in $\mathrm{SL}_2(\mathcal{O}_L)\backslash\mathbb{H}^6$. Thus the universe is the norm-π packet in this marked CM field, rather than a union over other CM fields. For an ideal $\mathfrak{f} \subseteq \mathcal{O}_L$, put $S_\mathfrak{f} = \mathcal{O}_L + \mathfrak{f}\mathcal{O}_M$; in particular $S_{\mathcal{O}_L} = \mathcal{O}_M$.

Lemma 3.1 (Point fibers and the intrinsic multiplier order)**.** *For any fixed trace c_k, the map from its integral matrix classes to quotient upper fixed points has exactly two classes in each fiber, represented by T and $c_kI - T$. The relative conductor ideal of the multiplier order is an invariant of the quotient point. In particular, different embedded ideals in Proposition* S.8.3 *give disjoint proper-multiplier point populations.*

Proof. The two-root argument in Theorem S.8.7 applies to every c_k: the characteristic polynomial is irreducible, since its discriminant $B_k^2(t^2 - 4)$ is negative at every real embedding. If two actions fix the same upper tuple, they preserve both conjugate complex lines and commute. Injectivity of a real embedding makes this commutation an identity over L. They therefore belong to the same quadratic centralizer, where their characteristic polynomial has only its two roots. An anti-intertwiner has determinant $-N_{M/L}(a)$, totally negative, so the two root classes are distinct under SL_2.

The L-algebra preserving those lines is intrinsic to the tuple. Its intersection with $M_2(\mathcal{O}_L)$ is the multiplier ring of the corresponding lattice. An integral determinant-one change of basis conjugates both the algebra and this order. The two identifications of that algebra with M fixing L differ only by complex conjugation, which preserves every $S_\mathfrak{f}$. Hence the exact ideal $\mathfrak{f}$, not merely its norm, is invariant. Extra endomorphisms of an unmarked abelian variety do not enlarge this two-by-two L-linear centralizer, and automorphisms permuting the marked real embeddings are not quotient identifications. □

Proposition 3.2 (The 144 selected $d = 3$ ambient points)**.** *The* 1280 *matrix classes of Theorem* S.8.6 *define exactly* 144 *distinct quotient points. Their decomposition and the number of those matrix classes incident to each point are*

Multiplier conductor	*Points*	*Incidences per point*
$\mathcal{O}_L$	16	24
$(7, t-1)$	32	16
$(7, t-1)^2$	32	4
$(43, t-20)$	32	4
$(43, t-31)$	32	4.

Their intersection with the 112*-point population of Theorem* S.8.7 *consists of exactly the sixteen maximal points; the union contains* 240 *distinct ambient points.*

Proof. Theorem S.8.6 supplies the complete module and orientation lists. Choose the common basis $(1, gz)$ with

$$g = 1,\ B_0,\ B_{12},\ B_4,\ B_8$$

for the five displayed multipliers. Let δ_m be the binary product selected by mask m from

$$(\xi_5, \xi_{11}, \xi_{13}, \xi_{17}, -1, u), \qquad u = 4 - t^2.$$

Use masks $0, \dots, 31$ for maximal multiplier and $0, \dots, 63$ for a proper multiplier. Multiplication by any allowed α_k in orientation $\delta = \delta_m$ is

$$T_{k,g,\delta} = \begin{pmatrix} A_k & -\delta g B_k \\ B_k/(\delta g) & A_k + tB_k \end{pmatrix}.$$

Putting $q = g\delta$, the upper tuple is specified exactly by

$$2x = -qt, \qquad 4h^2 = q^2(4 - t^2), \qquad |w|^2 = q^2, \qquad h > 0. \tag{3.1}$$

It is independent of the trace label. The matrix

$$U_q = \begin{pmatrix} 1 & qt \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathcal{O}_L)$$

sends this tuple to the one for $-q$ and satisfies $U_q(c_k I - T_{k,g,\delta})U_q^{-1} = T_{k,g,-\delta}$. Thus root pairing toggles bit 16. Canonical masks are $0, \dots, 15$ for $\mathcal{O}_M$ and $0, \dots, 15, 32, \dots, 47$ for every proper multiplier. Lemma 3.1 proves that no further quotient identifications occur. Literal equality of the chosen tuples is a separate assertion: their real parts force equality of q, whose ideal $(q) = (g)$ separates the five strata.

We also account for the basis orientations in the old 1280-matrix list. A regular basis $(1, \alpha_k)$ changes to $(1, gz)$ by $Q = \left(\begin{smallmatrix} 1 & A_k \\ 0 & e \end{smallmatrix}\right)$, where $e = B_k/g$; for the intermediate basis use $Q = \mathrm{diag}(1, (t-1)/B_0)$. The nontrivial unit-square identities $e = \delta_a v^2$ are

e	a	v
B_{15}/B_0	33	$1 + 5t - 5t^3 + t^5$
B_{18}/B_0	39	$3 + 13t - t^2 - 11t^3 + 2t^5$
$(t-1)/B_0$	7	$1 + 18t + 3t^2 - 16t^3 - t^4 + 3t^5$.

Each displayed v is a unit. For regular labels $0, 4, 8, 12$ use $e = v = 1, a = 0$; the opposite trace label negates e and toggles bit 16 of a. The intermediate basis is the same at labels 12 and 33. If $n = m \oplus a$ and $h_0 = \prod_{i:a_i=1,\, m_i=0} s_i$ for the six listed generators s_i, then $\delta_n \delta_a / \delta_m = h_0^2$. Consequently

$$P = (vh_0)^{-1} D_{\delta_n} Q D_{\delta_m}^{-1}, \qquad D_\delta = \mathrm{diag}(\delta, 1),$$

is integral of determinant one and transports the old action to its common-basis action. If needed, compose with $U_{g\delta_n}$ to select the canonical root partner. The paired exact implementations in Supplement, Section S.13 check every unit identity and all 1280 matrix and upper-tuple transports.

There are twelve trace labels at maximal multiplier, eight at multiplier $(7, t-1)$, and two at each of the other three proper multipliers. The two-root fibers give the displayed incidences and $16 \cdot 24 + 32 \cdot 16 + 96 \cdot 4 = 1280$. The intermediate conductor-seven stratum is counted once. The proper $d = 3$ conductors differ from the two proper $d = 7$ conductors, whereas their common maximal tuples coincide. Lemma 3.1 gives the asserted intersection and $144 + 112 - 16 = 240$. □

Theorem 3.3 (The full marked norm-π arithmetic census)**.** *For all* 42 *traces in Proposition* S.8.3*, the complete integral matrix universe has* 3520 *trace-labelled* $\mathrm{SL}_2(\mathcal{O}_L)$ *classes. The exact counts, including every module overorder and determinant orientation, are*

Generated-order conductor	*Traces*	*Classes per trace*	*Subtotal*
$\mathcal{O}_L$	16	32	512
$(7, t-1)$	12	96	1152
$(7, t-1)^2$	2	160	320
$(41, t-12)$	2	96	192
$(41, t-22)$	2	96	192
$(43, t-20)$	2	96	192
$(43, t-31)$	2	96	192
$(83, t-36)$	2	160	320
$(127, t-94)$	2	224	448.

Their upper fixed tuples define exactly 368 *distinct ambient quotient points, with the following intrinsic multiplier decomposition:*

Multiplier conductor	$\lvert \mathrm{Pic}(S_{\mathfrak{f}}) \rvert$	*Points*	*Incidences per point*
$\mathcal{O}_L$	1	16	84
$(7, t-1)$	1	32	28
$(7, t-1)^2$	1	32	4
$(41, t-12)$	1	32	4
$(41, t-22)$	1	32	4
$(43, t-20)$	1	32	4
$(43, t-31)$	1	32	4
$(83, t-36)$	2	64	4
$(127, t-94)$	3	96	4.

In particular the two nonprincipal classes at conductor $(127, t-94)$ *are exchanged by conjugation and together contribute* 64 *points. These are arithmetic ambient points; the theorem makes no new curve-membership or geometric CM-type assertion.*

Proof. The existing local calculations in Theorems S.8.5 and S.8.6 leave two embedded orders to consider. Put

$$\mathfrak{p}'_{41} = (41, t-22), \quad \mathfrak{p}_{127} = (127, t-94), \quad \gamma = B_{10}.$$

Proposition S.8.3 gives $(B_5) = \mathfrak{p}'_{41}$ and $(\gamma) = \mathfrak{p}_{127}$. For either prime $\mathfrak{p} = (p, t - r)$, compute in

$$A_\mathfrak{p} = \mathbf{F}_p[Z]/(Z^2 - rZ + 1), \qquad Q_\mathfrak{p} = A_\mathfrak{p}^\times/\mathbf{F}_p^\times.$$

Exact multiplication of projective pairs $[a : b]$, with $a^2 + rab + b^2 \neq 0$, gives

$\mathfrak{p}$	$A_\mathfrak{p}$	$\lvert Q_\mathfrak{p}\rvert$	$[\varepsilon]$	ord$[\varepsilon]$
$\mathfrak{p}'_{41}$	$\mathbf{F}_{41^2}$	42	$[1 : 30]$	42
$\mathfrak{p}_{127}$	$\mathbf{F}_{127} \times \mathbf{F}_{127}$	126	$[1 : 80]$	42.

Here $\varepsilon = 1 - z^2$ and $[z] = [\varepsilon]^{22}$ in both quotients. At 127 the roots are $40, 54$; the ordered ratio of the two components sends ε to 51 and $1 + 9z$ to 7. For example,

$$51^6 = 32, \quad 51^{14} = 107, \quad 51^{21} = -1, \quad 51^{42} = 1,$$
$$7^{18} = 64, \quad 7^{42} = 107, \quad 7^{63} = -1, \quad 7^{126} = 1 \pmod{127}$$

certify their respective orders 42 and 126. The complete unit generation $E_M = \langle\varepsilon\rangle\mu_{42}E_L$ proved in Theorem S.8.5 shows that no omitted global unit enlarges the order-42 image. The conductor exact sequence and $h(M) = 1$ give

$$\mathrm{Pic}(S_{\mathfrak{p}'_{41}}) = 1, \qquad \mathrm{Pic}(S_{\mathfrak{p}_{127}}) \simeq \mathbf{Z}/3\mathbf{Z}.$$

For every proper order here, the unit condition is $n + 22j \equiv 0 \pmod{d}$ on $\varepsilon^n z^j v$, $v \in E_L$, where the image order d is 6 or 42. At 127 it is 42, not $|Q_{\mathfrak{p}_{127}}| = 126$. The exponent n is therefore even, so $N(S_\mathfrak{f}^\times) = E_L^2$; conversely all real unit squares occur. For maximal multiplier the image is E_L^+.

For completeness of the modules at a prime conductor, normalize a full local rank-one lattice I so its maximal-order span is B. In the inert case a vector of minimal valuation is a unit of B; in the split case the residue subspace surjects to each coordinate and contains a vector nonzero in both. Divide by such a unit. Then $D + \varpi B \subseteq I \subseteq B$, whose quotient has length one over the local DVR D. Hence there are exactly the proper and maximal local types. Globally this gives

$$\mathrm{ICM}(S_\mathfrak{p}) = \mathrm{Pic}(S_\mathfrak{p}) \sqcup \mathrm{Pic}(\mathcal{O}_M).$$

The square-conductor proof in Theorem S.8.6 adds exactly the intermediate order at seven. Thus noninvertible modules over the originally generated order are included. Since $h(L) = 1$, each lattice has an integral rank-two basis. The determinant quotient $E_L/N(S^\times)$ supplies 32 classes for maximal multiplier and 64 for each proper Picard class. These facts prove the first table.

To make the new nonprincipal classes and their root fibers explicit, use the three distinct conductor-unit cosets $1, 1 + 9z, 1 + 100z$ at 127 and the ideals

$$J_s = \gamma\mathcal{O}_M + (1 + sz)\mathcal{O}_L, \qquad (1 + sz, \gamma z) \text{ a basis}, \qquad s = 0, 9, 100.$$

Their residues are units at the conductor; locally they are multiples of the proper order, and away from it they are maximal. Thus they are invertible ideals of the proper multiplier order. The conductor sequence identifies their three classes and proves they exhaust it. The action of $\alpha_k = A_k + B_k z$ is

$$T_{k,s} = \begin{pmatrix} A_k - sB_k & -\gamma B_k \\ (B_k/\gamma)(1 + st + s^2) & A_k + (t + s)B_k \end{pmatrix}.$$

Its entries are integral and its characteristic polynomial is the required one; conjugating by D_{δ_m} supplies all orientations.

Conjugation is inversion on $Q_{\mathfrak{p}_{127}}$, and $\overline{1+9z} = 85(1+100z)$ modulo the conductor. Consequently $\overline{J_9} = J_{100}$ and conversely. The conjugation matrix from the basis of J_9 to that of J_{100} is

$$C_{100,9} = \begin{pmatrix} 1+9t & \gamma t \\ -(109+900t)/\gamma & -1-100t \end{pmatrix}, \qquad \det C_{100,9} = -1.$$

It is integral, since $109+900\cdot 94 \equiv 0 \pmod{127}$ and $(\gamma) = (127, t-94)$. Therefore $D_{\delta_{m\oplus 16}} C_{100,9} D_{\delta_m}^{-1}$ has determinant one and transports the action on J_9 to the conjugate-root action on J_{100}. The root pairs are $(J_9, m) \leftrightarrow (J_{100}, m \oplus 16)$. All 64 masks on J_9 represent the nonprincipal point population; the principal class contributes 32 further points, giving 96.

For clarity, the nonprincipal conductor-83 module may be represented by $(1+3z, B_2 z)$. In its conductor quotient $[1+3z]^2 = [\varepsilon]^{29}$, and $\eta = \varepsilon z^{28}$ has norm u. Multiplication by η followed by conjugation preserves this ideal and has determinant $-u$. Thus its root pairing is $m \mapsto m \oplus 48$. After twisting, multiply the bridge by the scalar u^{-1} when bit 32 of m is unset, and by 1 otherwise; the determinant is then one. Principal modules use $m \mapsto m \oplus 16$. The two portable implementations check all 736 integral root bridges against every permitted trace action.

Finally, in any basis $(1+sz, gz)$ put $q = g\delta$ and $n = 1+st+s^2 = N_{M/L}(1+sz) > 0$. The exact tuple satisfies

$$2x = -\frac{q(t+2s)}{n}, \qquad 4h^2 = \frac{q^2(4-t^2)}{n^2}, \qquad |w|^2 = \frac{q^2}{n}.$$

It is independent of the allowed trace label. Together with Lemma 3.1, this identifies repeated overorders and proves both the point counts and separation of distinct conductor strata in the second table. The fourteen traces acting in the conductor-seven order give 28 incidences per point; all 42 act in the maximal order. Each other proper stratum occurs for two traces. Thus the independent incidence check is

$$16\cdot 84 + 32\cdot 28 + 320\cdot 4 = 3520, \qquad 16 + 6\cdot 32 + 64 + 96 = 368.$$

The exact manifests also compare all chosen tuple coefficients, but quotient distinctness follows from the structural arguments above. □

Remark 3.4 (Geometry and representative labels). The 144-point and 112-point populations account for 240 points of the full census. The two further conductor strata contribute $32 + 96 = 128$ points, giving 368. Equal rational conductor norms do not identify the two primes over 41 or the two over 43. The common-basis labels used here need not equal older companion or ideal-lattice masks, especially at nonprincipal conductor 83. An individual geometric decision requires an exact representative transport; a proved whole-stratum conclusion transfers through the equality of complete intrinsic populations. This arithmetic census adds no geometric CM type, polarization or Hodge conclusion.

3.2 From two regions to the full curve

Lemma 3.5 (Two simultaneous target regions)**.** *Let Q be the labelled quadrilateral chamber in McMullen's source half-plane, let r be reflection in its AB side, and let Q_i be its six normalized coordinate quadrilaterals. Write r_i for reflection in the labelled A_iB_i side and put*

$$P_0 = \prod_{i=1}^{6} Q_i, \qquad P_1 = \prod_{i=1}^{6} r_i(Q_i).$$

If a tuple $p \in \mathbb{H}^6$ represents a point of the normalized McMullen curve, some $H \in \mathrm{SL}_2(\mathcal{O}_L)$ sends p into $P_0 \cup P_1$. Consequently, complete empty searches for both target regions exclude p from the curve. Each search may use its own explicitly verified $\mathrm{SL}_2(\mathcal{O}_L)$ translate of p.

Proof. Let W be the quadrilateral reflection group and let Δ_0 be its orientation-preserving subgroup. McMullen's construction identifies the holomorphic quadrilateral group with this index-two subgroup [6, §4]. Thus $Q \cup rQ$ is a fundamental region for Δ_0, with boundary identifications allowed. The normalized modular embedding F is obtained from the labelled triangle maps by Schwarz continuation. The same continuation across the AB side gives $F_i(rz) = r_i(F_i(z))$ in every coordinate [6, §6, Theorem 6.3 and the proof of Theorem 1.9]. It follows that $F(Q) \subset P_0$ and $F(rQ) \subset P_1$. The region choice is simultaneous in all six coordinates; no independent coordinate reflections are used.

Write $p = KF(z)$ with $K \in \mathrm{SL}_2(\mathcal{O}_L)$. Choose $\eta \in \Delta_0$ such that $\eta z \in Q \cup rQ$. Equivariance gives $\rho(\eta)K^{-1}p = F(\eta z) \in P_0 \cup P_1$, where the normalized monodromy lift $\rho(\eta)$ is integral of determinant one. If search j uses $p_j = G_j p$ with $G_j \in \mathrm{SL}_2(\mathcal{O}_L)$, then HG_j^{-1} carries p_j to the same target. Both empty exhaustive searches contradict this necessity. This proof makes no assertion that any r_i normalizes the arithmetic group or permutes the packet. □

Theorem 3.6 (Complete fixed marked geometric packet)**.** *Assume the packet execution hypothesis stated in the introduction. Let $\mathcal{S}$ be the* 368 *ambient quotient points of Theorem 3.3, with its fixed M/L, norm generator, L-action and ordered embeddings. Let $\mathcal{X}$ be the normalized McMullen curve image in $\mathrm{SL}_2(\mathcal{O}_L)\backslash\mathbb{H}^6$. Then*

$$\mathcal{S} \cap \mathcal{X} = \{A, B, C\}.$$

Thus exactly 365 *points are excluded, including every point of proper multiplier order. The three labels in the arithmetic certificate are* `OM:pic0:07`*,* `OM:pic0:15` *and* `OM:pic0:00` *for A, B, C, respectively.*

Proof. The arithmetic census fixes the complete domain, including all module classes, determinant sectors, intermediate overorders and root fibers. The positive vertex realizations in Theorem 1.2 and Theorem S.8.7 give three distinct points in the intersection. It remains to exclude each of the other 365 points.

For the original target P_0, the retained legacy calculations cover 116 distinct points. The original-target portable certificate covers the disjoint complement of 249 nonvertex labels. Their union is exactly $\mathcal{S} \setminus \{A, B, C\}$; the thirteen maximal exclusions are already inside the 116 and are not counted twice. The original-target coverage uses the complete numerical searches, not the full-curve conclusions of the historical subpacket propositions. At nonprincipal conductor 83, its bridge is equality of the entire 32-point population, not an unproved permutation of the old and new mask labels.

The reflected certificate reconstructs all 365 nonvertex labels and tests the simultaneous target P_1. In each target the source transport is separately verified integral of determinant one. Fresh exact target vertices and outward rational height/modulus bounds give finite positive trace-form ellipsoids for all possible entries. Inclusive lattice recursion, the affine determinant equation, both simultaneous matrix signs and the separate $c = 0$ branch exhaust the candidate set. Closed side predicates retain boundary equality. These arguments apply to nonprincipal modules and all orientations, without a maximal-order or mask-zero hypothesis.

The primary quotient-field path and the separate Sturm path agree on the complete preparations, entries, rows, candidates and classifications. Every survivor set is empty. The retained reports bind each plan and result to its source, target, runtime and full row journal. The whole-executable correspondence and actual portable controls justify the source relocation; controls are

not a sampling proof for untested labels. The certificates and historical execution assumptions are specified in Supplement, Section S.13.

If a remaining point lay on the curve, Lemma 3.5 would give an integral determinant-one matrix sending it into P_0 or P_1. For a source replaced by $G_j p$ in target j, the corresponding matrix is HG_j^{-1}. The complete search for that target would retain it, contradicting its empty result. The positive realizations therefore give the asserted equality. No arithmetic-normalizer property of the reflection and no independent per-coordinate chamber choice is used. □

4 The explicit polarization on the normalized Hilbert modular model

The preceding arguments use $X_L = \mathbb{H}^6/\mathrm{SL}_2(\mathcal{O}_L)$ as an analytic Hilbert modular quotient. To apply a theorem about polarized abelian varieties we specify the polarization directly rather than infer principal polarization from the notation X_L.

For $z = (z_1, \dots, z_6) \in \mathbb{H}^6$, let $\sigma_1, \dots, \sigma_6$ be the real embeddings of L and set

$$\Lambda_z = \{(\sigma_i(a)z_i + \sigma_i(b))_{i=1}^6 : a, b \in \mathcal{O}_L\} \subset \mathbb{C}^6, \qquad A_z = \mathbb{C}^6/\Lambda_z.$$

On the abstract lattice $\mathcal{O}_L^2$ define

$$E_1((a,b),(a',b')) = \mathrm{Tr}_{L/\mathbb{Q}}(a'b - ab'). \tag{4.1}$$

This form is integral and alternating. With the displayed sign convention, at one real place, if $v = (a,b)$ and J_z is multiplication by i in the coordinate $az + b$, then

$$E_1(v, J_z v) = a^2\ \mathrm{Im}(z) + \frac{(a\,\mathrm{Re}(z) + b)^2}{\mathrm{Im}(z)} > 0$$

for $v \neq 0$. Summing over the six embeddings verifies the Riemann bilinear positivity criterion, so E_1 is a polarization. The $\mathcal{O}_L$-action is self-adjoint for E_1, and every matrix in $\mathrm{SL}_2(\mathcal{O}_L)$ preserves (4.1). Thus this construction descends to the quotient stack $[\mathbb{H}^6/\mathrm{SL}_2(\mathcal{O}_L)]$ (equivalently, to the Hilbert modular orbifold X_L) as a polarized RM family. After passage to any neat finite-index subgroup it becomes an honest abelian scheme over the corresponding finite cover. In the arguments below only the explicit fiber at z_B is needed.

Proposition 4.1 (Polarization type)**.** *For the integral basis $1, t, \dots, t^5$ of $\mathcal{O}_L$, the polarization E_1 has type*

$$\boxed{(1,7,7,21,21,21)}.$$

In particular, the polarization used below is not principal. Its degree is

$$(1 \cdot 7 \cdot 7 \cdot 21 \cdot 21 \cdot 21)^2 = \mathrm{disc}(L)^2.$$

Proof. Let

$$T = (\mathrm{Tr}_{L/\mathbb{Q}}(t^{i+j}))_{0 \le i,j \le 5}.$$

In the induced basis of $\mathcal{O}_L^2$, the alternating matrix of E_1 is $\left(\begin{smallmatrix} 0 & -T \\ T & 0 \end{smallmatrix}\right)$. Exact Smith reduction gives

$$\mathrm{SNF}(T) = \mathrm{diag}(1,7,7,21,21,21), \qquad \det T = 3^3 7^5 = 453789 = \mathrm{disc}(L).$$

The elementary divisors of the alternating form are therefore the six displayed entries, and the polarization degree is the square of their product. □

Proposition 4.2 (No compatible principal polarization on the explicit $\mathcal{O}_L^2$ lattice)**.** *For the fixed real-multiplication structure on $H_1(A_z, \mathbb{Z}) \cong \mathcal{O}_L^2$, there is no principal polarization whose Rosati involution fixes L pointwise.*

Proof. An L-self-adjoint alternating form on the rank-two L-space L^2 is necessarily of the form

$$E_\lambda((a,b),(a',b')) = \mathrm{Tr}_{L/\mathbb{Q}}(\lambda(a'b - ab'))$$

for a unique $\lambda \in L$. On the lattice $\mathcal{O}_L^2$ it is integral exactly when $\lambda \in \mathfrak{D}_L^{-1}$, and it is unimodular exactly when $\lambda\mathfrak{D}_L = \mathcal{O}_L$. With $z \in \mathbb{H}^6$, the Riemann positivity condition is $\sigma_i(\lambda) > 0$ for all six real embeddings. Thus a compatible principal polarization exists precisely when the codifferent has a totally positive generator.

Because $\mathcal{O}_L = \mathbb{Z}[t]$ with minimal polynomial f, the different is $(f'(t))$. In the embedding order $k = 1, 5, 11, 13, 17, 19$, the signs of $f'(\sigma_k(t))$ are

$$(+,-,+,-,+,-).$$

Supplement, Section S.8 computes the image of the unit-sign map $E_L \to \{\pm 1\}^6$: it is the rank-five subgroup generated by the sign vectors of $\xi_5, \xi_{11}, \xi_{13}, \xi_{17}, \xi_{19}$ and -1. Exact row reduction over $\mathbf{F}_2$ shows that $(+,-,+,-,+,-)$ is not in this subgroup. Hence no unit multiple of $1/f'(t)$ is totally positive, so $\mathfrak{D}_L^{-1}$ has no totally positive generator. □

Homology convention. The column (a,b) represents $az+b$. If a Mobius matrix $M = \left(\begin{smallmatrix} r & s \\ u & v \end{smallmatrix}\right)$ fixes z, then multiplication by $uz+v$ on these columns is M^{T}, as proved in Proposition 5.4. Accordingly, a Mobius-polynomial representative S_X of $\sqrt{-3}$ acts on homology as $\Sigma_X = S_X^{\mathsf{T}}$. All lattice stability and Hermitian forms below use Σ_X.

Proposition 4.3 (A principally polarized model in the isogeny class)**.** *Let $\Lambda = \mathcal{O}_L^2$ be the homology lattice of A_B, let*

$$\Lambda^\# = \{v \in L^2 : E_1(v, \Lambda) \subset \mathbb{Z}\} = \mathfrak{D}_L^{-1}\Lambda,$$

and let $\Sigma_B = (I + 2B_t^{14})^{\mathsf{T}}$. There is an $\mathcal{O}_L$-submodule

$$H \subset \Lambda^\#/\Lambda$$

which is maximal isotropic for the discriminant pairing and stable under the action of $\mathcal{O}_K$. Here $\sqrt{-3}$ acts as Σ_B, while the integral generator

$$U := (B_t^{14})^{\mathsf{T}} = \frac{\Sigma_B - I}{2}$$

satisfies $U^2 + U + I = 0$ and hence identifies $\mathcal{O}_K$ with $\mathbb{Z}[U]$. If Λ^{pp} denotes the inverse image of H in $\Lambda^\#$, then

$$[\Lambda^{\mathrm{pp}} : \Lambda] = \mathrm{disc}(L) = 453789, \qquad (\Lambda^{\mathrm{pp}})^\# = \Lambda^{\mathrm{pp}}.$$

Consequently

$$A_B^{\mathrm{pp}} := \mathbb{C}^6/\Lambda_B^{\mathrm{pp}}$$

is a principally polarized abelian sixfold, the natural map

$$A_B \longrightarrow A_B^{\mathrm{pp}}$$

is an $\mathcal{O}_L$-linear isogeny of degree 453789, and the K-action descends to A_B^{pp}. The rational Hodge structure, K-signature $(3,3)$, Weil discriminant -1, and validity of the Hodge conjecture are unchanged under this isogeny.

Proof. Since $\mathcal{O}_L = \mathbb{Z}[t]$, Dedekind factorization applied to (2.1) gives

$$f(T) \equiv (T^3 - T^2 + T + 1)^2 \pmod 3, \qquad f(T) \equiv (T-1)^6 \pmod 7.$$

Both ramification indices are prime to the residue characteristic, hence the ramification is tame. Thus

$$(3) = \mathfrak{p}_3^2, \quad N\mathfrak{p}_3 = 3^3, \qquad (7) = \mathfrak{p}_7^6, \quad N\mathfrak{p}_7 = 7,$$

and the different is

$$\mathfrak{D}_L = \mathfrak{p}_3\mathfrak{p}_7^5.$$

Multiplication by a generator of the different identifies the discriminant module with

$$\Lambda^{\#}/\Lambda \cong (\mathcal{O}_L/\mathfrak{D}_L)^2 \cong (\mathcal{O}_L/\mathfrak{p}_3)^2 \oplus (\mathcal{O}_L/\mathfrak{p}_7^5)^2.$$

At $\mathfrak{p}_3$, exact reduction of the displayed matrix Σ_B gives

$$\Sigma_B \equiv 0 \pmod{\mathfrak{p}_3}.$$

Hence any one-dimensional $\mathcal{O}_L/\mathfrak{p}_3 \cong \mathbb{F}_{27}$ subspace is Σ_B-stable; take the first-coordinate line H_3. Since $U = (\Sigma_B - I)/2 \equiv 1 \pmod{\mathfrak{p}_3}$, this line is already $\mathcal{O}_K = \mathbb{Z}[U]$-stable.

At $\mathfrak{p}_7^5$, the quotient has characteristic 7 and $\Sigma_B^2 = -3I = 4I$. Moreover $U^2 + U + I = 0$ gives $\operatorname{tr}(U) = -1$, hence

$$\operatorname{tr}(\Sigma_B) = \operatorname{tr}(I + 2U) = 0.$$

Since 4 is a unit, the idempotents

$$e_+ = \frac{\Sigma_B + 2I}{4}, \qquad e_- = \frac{2I - \Sigma_B}{4}$$

split $(\mathcal{O}_L/\mathfrak{p}_7^5)^2$ into the $+2$ and -2 eigensummands. Because the local ring $\mathcal{O}_L/\mathfrak{p}_7^5$ is local, the images of these idempotents are finite projective and hence free. Moreover

$$\operatorname{tr}(e_+) = \frac{\operatorname{tr}(\Sigma_B) + 4}{4} = 1, \qquad \operatorname{tr}(e_-) = 1,$$

so both images have rank one. Let $H_7 = \operatorname{im}(e_+)$. Because $\Sigma_B^{\dagger} = -\Sigma_B$, for $v, w \in H_7$ one has

$$0 = E_1(\Sigma_B v, w) + E_1(v, \Sigma_B w) = 4E_1(v, w)$$

in the local discriminant pairing, so H_7 is isotropic. The line H_3 is isotropic for the same alternating pairing. By the Chinese remainder theorem

$$H = H_3 \oplus H_7$$

is isotropic and $\mathcal{O}_L$-stable, and

$$|H| = 3^3\,7^5 = 453789 = \operatorname{disc}(L) = \sqrt{|\Lambda^{\#}/\Lambda|}.$$

Thus H is maximal isotropic and its inverse image Λ^{pp} is self-dual for E_1.

The action of Σ_B preserves H. The full ring $\mathcal{O}_K = \mathbb{Z}[U]$ preserves it as well. Indeed $U = (B_t^{14})^{\mathsf{T}}$ preserves both Λ and $\Lambda^{\#}$; on the discriminant module one also has $U = (\Sigma_B - I)/2$, and multiplication by 2 is invertible on both the 3- and 7-primary parts, so Σ_B-stability of H implies U-stability. Hence Λ^{pp} is $\mathcal{O}_K$-stable. The complex structure, the rational K-action on

$V := \Lambda \otimes \mathbb{Q} = L^2$, and the rational alternating form E_1 have not changed. Hence the K-Hermitian form $H(v,w) = E_1(\Sigma_B v, w) + \sqrt{-3}\, E_1(v,w)$ on the rational K-space V is literally the same form before and after the lattice enlargement. In particular its determinant class in $\mathbb{Q}^\times / N_{K/\mathbb{Q}}(K^\times)$ remains -1, not merely up to an appeal to isogeny invariance. The CM type and all rational Hodge classes are likewise unchanged. The inclusion $\Lambda \subset \Lambda^{\mathrm{pp}}$ has index $|H|$, which is the degree of the induced isogeny. Finally the Hodge conjecture is invariant under isogeny (pullback and pushforward along a finite isogeny preserve algebraicity and induce inverse isomorphisms on rational cohomology up to the degree). □

Corollary 4.4 (The principal model is not the same orbifold vertex)**.** *No principally polarized self-dual overlattice*

$$\Lambda \subset \Lambda' \subset \Lambda^{\#}$$

can be stable under the full vertex automorphism acting as B_t^{T} on homology. In particular the principally polarized model of Proposition 4.3 cannot be obtained by merely changing the polarization while retaining the same integral order-42 B_t-vertex structure.

Proof. Every self-dual overlattice corresponds to a maximal isotropic subgroup, whose $\mathfrak{p}_3$-part has order 27. Stability under the actual homology automorphism B_t^{T} implies stability under its inverse and therefore under their sum tI. Hence this subgroup is an $\mathbb{F}_{27}$-line in $(\mathcal{O}_L/\mathfrak{p}_3)^2$. The characteristic polynomial of B_t is $X^2 - tX + 1$. Its discriminant $t^2 - 4$ is a nonsquare in $\mathbb{F}_{27}$: exact reduction modulo $g_3(T) = T^3 - T^2 + T + 1$ gives

$$(t^2 - 4)^{13} = -1.$$

Hence $X^2 - tX + 1$ is irreducible over $\mathbb{F}_{27}$ and B_t has no invariant $\mathbb{F}_{27}$-line. Thus no maximal isotropic subgroup, and hence no principally polarized overlattice, can be B_t^{T}-stable. □

Remark 4.5 (From special fibers to a common principal family). Proposition 6.1 below supplies a common nonfree maximal isotropic subgroup for the three specialized K-actions. Proposition 6.3 determines its ambient index 224 and its index 28 in the explicitly parametrizing quadrilateral group, before auxiliary level. Theorem 6.4 descends its inverse-image lattice sheaf and principal polarization over a connected neat cover. Proposition 6.5 determines the surviving order inside each selected CM field. The value 537824 in the earlier free-line construction concerns a different subgroup and is not the index of this common nonfree level.

Remark 4.6 (What the principally polarized model does not prove). The special-fiber subgroup H gives a principally polarized model A_B^{pp}; Remark 4.5 describes the later common choice used for the family. Corollary 4.4 shows that any such modification necessarily loses the integral order-42 vertex automorphism at the prime above 3, so it would not be the original orbifold B-vertex in another polarization disguise. We do not identify A_B^{pp} with a point on Mostaed's original curve V inside the principally polarized Hilbert–Blumenthal moduli space.

Thus the Hodge-theoretic conclusion below does not require a map from X_L to the principally polarized moduli space $\mathcal{A}_6$: it uses the explicit polarized abelian variety (A_z, E_1) at the certified vertex. In fact Proposition 4.2 shows that, for this fixed $\mathcal{O}_L$-module and real-multiplication action, the missing principal polarization is not merely unconstructed. Markman's theorem is stated for polarized abelian sixfolds, not only principally polarized ones.

At the normalized B-vertex put

$$U = (B_t^{14})^{\mathsf{T}}, \qquad \Sigma_B = I + 2U.$$

Since $B_t^{\mathsf{T}} \in \mathrm{SL}_2(\mathcal{O}_L)$, its E_1-adjoint is $(B_t^{-1})^{\mathsf{T}}$. The elliptic relation gives $U^2 + U + I = 0$, and hence

$$\Sigma_B^2 = -3I, \qquad \Sigma_B^\dagger = I + 2U^{-1} = I + 2U^2 = -\Sigma_B, \qquad \Sigma_B^\dagger \Sigma_B = 3I.$$

Proposition 4.7 (Polarization-compatible K-action)**.** *The assignment $\sqrt{-3} \mapsto \Sigma_B$ defines an embedding*

$$K = \mathbb{Q}(\sqrt{-3}) \hookrightarrow \mathrm{End}^0(A_z)$$

at the certified B_t-vertex and is compatible with E_1: for every $k \in K$,

$$\eta(k)^\dagger \eta(k) = N_{K/\mathbb{Q}}(k) I.$$

Proof. The identity $\Sigma_B^2 = -3I$ gives the field embedding. Since the vertex is fixed by B_t, the actual homology action is B_t^{T}; every polynomial in it, in particular Σ_B, preserves the complex structure. Integrality gives an actual endomorphism. The relation $\Sigma_B^\dagger = -\Sigma_B$ gives

$$(a - b\Sigma_B)(a + b\Sigma_B) = a^2 + 3b^2 = N_{K/\mathbb{Q}}(a + b\sqrt{-3}),$$

and the claim follows for all $a, b \in \mathbb{Q}$. □

5 The three geometric fibers

We now supply the positive realizations and Hodge data used above. The B companion bridge fixes the geometric marking explicitly. The A and C calculations below use their actual vertex actions; their CM types are not inferred from an arithmetic companion signature.

5.1 The arithmetic certificate used in Theorem 1.2

Only a small part of the $\ell = 43$ arithmetic is needed for the main theorem. The full norm-fiber classification, signature enumeration, conjugacy reduction, component analysis, and unit calculation are deferred to Supplement, Section S.2.

Put

$$\pi = t^4 + t^3 - 4t^2 - 2t + 5, \qquad \alpha_0 = 1 + \zeta_{42} + \zeta_{42}^4, \qquad \alpha_{11} = \zeta_{42}^{11} \alpha_0,$$

and $c_{11} = \alpha_{11} + \overline{\alpha}_{11}$. Define

$$A_{c_{11}} = \begin{pmatrix} 0 & -\pi \\ 1 & c_{11} \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -1 \\ 1 & t \end{pmatrix}.$$

Proposition 5.1 (Arithmetic and component certificate)**.** *The following assertions hold.*

(i) $N_{M/L}(\alpha_0) = \pi$, $N_{L/\mathbb{Q}}(\pi) = 43$, *and π is totally positive. Thus $A_{c_{11}}$ is an integral determinant-π Hecke matrix with characteristic polynomial $X^2 - c_{11}X + \pi$.*

(ii) The six Galois triangles in McMullen's construction have orientations, relative to the identity component and ordered by $k = 1, 5, 11, 13, 17, 19$,

$$(+, +, -, -, -, -).$$

The unit t has the same sign vector. Hence

$$h_t = \mathrm{diag}(t, 1), \qquad B_t = h_t B h_t^{-1}$$

changes all six components to $\mathbb{H}$ and normalizes $\mathrm{SL}_2(\mathcal{O}_L)$.

(iii) The point defined by $A_{c_{11}}$ belongs to the maximal-order branch $\mathcal{O}_L[\alpha_{11}] = \mathcal{O}_M$; no additional integral-conjugacy or unit-norm ambiguity is required for the exact certificate of Section 5.2.

Proof. Part (i) is the exact cyclotomic calculation of Supplement, Section S.3. For (ii), McMullen constructs the component maps by comparing the orientations of the six Galois triangles and sends an oppositely oriented component to $-\mathbb{H}$ while preserving the equivariance relation; see [6, Prop. 6.1, Cor. 6.2, Thm. 6.3 and Eq. (6.1)]. The six signs are certified in Supplement, Section S.7; $N_{L/\mathbb{Q}}(t) = 1$, so $t \in \mathcal{O}_L^\times$ and conjugation by h_t preserves $\mathrm{SL}_2(\mathcal{O}_L)$. Part (iii) follows from the exact order-discriminant and unit-norm calculation in Supplement, Section S.8. □

This proposition is deliberately a certificate, not a reconstruction of the entire search. The proof of Theorem 1.2 from this point onward uses only the explicit matrices and the polarized Hodge structure at their common vertex.

Proposition 5.2 (Geometric meaning of the normalized B-vertex)**.** *Let $F : \mathbb{H} \to (\pm\mathbb{H})^6$ be McMullen's modular embedding constructed from the six Galois triangles, with the identity embedding as the first component. Let $z_B \in \mathbb{H}$ be the vertex fixed by the order-42 generator B in the identity triangle and put*

$$F_t = h_t \circ F, \qquad h_t = \mathrm{diag}(t, 1).$$

Then F_t takes values in $\mathbb{H}^6$, is equivariant for the conjugated integral representation $g \mapsto h_t g h_t^{-1}$, and $F_t(z_B)$ is the componentwise upper-half-plane fixed point of

$$B_t = h_t B h_t^{-1}.$$

Consequently the image of $F_t(z_B)$ in X_L is a point of the normalized McMullen geodesic, independently of any Hecke calculation.

Proof. McMullen's Theorem 6.3 constructs each component f_i by the unique vertex-respecting conformal map from the identity triangle to the i-th Galois triangle. When the two orientations agree, f_i takes values in $\mathbb{H}$; when they disagree, it takes values in $-\mathbb{H}$, and in both cases

$$f_i(g_1 z) = g_i f_i(z)$$

for every triangle-group element g [6, Thm. 6.3, Eq. (6.1)]. In particular $f_i(z_B)$ is the vertex fixed by B_i. Proposition 5.1 gives the component signs $(+, +, -, -, -, -)$ and the identical sign vector for the real determinants $\sigma_i(t)$ of h_t. Since

$$\mathrm{Im}(h_{t,i} w) = \frac{\sigma_i(t)\,\mathrm{Im}(w)}{|c_i w + d_i|^2}$$

for a real fractional-linear map of determinant $\sigma_i(t)$, h_t carries the raw component of F to $\mathbb{H}^6$. As $t \in \mathcal{O}_L^\times$, conjugation by h_t preserves $\mathrm{SL}_2(\mathcal{O}_L)$. Finally, conjugating the equation $B_i f_i(z_B) = f_i(z_B)$ by $h_{t,i}$ shows that the i-th coordinate of $F_t(z_B)$ is fixed by $(B_t)_i$. □

5.2 An exact $k = 11$ vertex intersection

The finite orbit search described above contains a particularly rigid case. It is not merely a numerical near-hit: after the integral component normalization $h_t = \mathrm{diag}(t, 1)$, one of the $k = 11$ Hecke fixed points is exactly the elliptic B-vertex of McMullen's normalized fundamental quadrilateral.

Let

$$B = \begin{pmatrix} 0 & -1 \\ 1 & t \end{pmatrix}, \qquad B_t = h_t B h_t^{-1},$$

and let $A_{c_{11}} = \left(\begin{smallmatrix} 0 & -\pi \\ 1 & c_{11} \end{smallmatrix}\right)$ be the companion representative associated with $\alpha_{11} = \zeta_{42}^{11}\alpha_0$. Define

$$G = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}$$

with

$$\begin{aligned} g_{11} &= 2t^5 - t^4 - 9t^3 + 3t^2 + 7t + 1, \\ g_{12} &= (t^2 - t - 1)(t^3 - 4t - 1), \\ g_{21} &= -2(t^5 - 5t^3 - t^2 + 5t + 3), \\ g_{22} &= -(t - 2)(3t^4 + 7t^3 - 4t^2 - 16t - 7). \end{aligned}$$

Theorem 5.3 (Exact B_t-vertex certificate)**.** *In $M_2(L)$ one has*

$$\det G = 1, \qquad [GA_{c_{11}}G^{-1}, B_t] = 0.$$

More precisely,

$$GA_{c_{11}}G^{-1} = xI + yB_t$$

with

$$x = (t^2 - 2)(t^3 - t^2 - 6t - 2), \qquad y = t(t^2 - 2)(t^2 + t - 1).$$

Consequently the Hecke fixed point represented by $A_{c_{11}}$ is $\mathrm{SL}_2(\mathcal{O}_L)$-equivalent to the geometrically defined point $F_t(z_B)$ of Proposition 5.2. In particular the corresponding quotient point lies on the normalized McMullen curve in X_L.

Proof. All identities are polynomial identities in $L = \mathbb{Q}[t]/(f)$. Direct reduction modulo f gives $\det G = 1$ and the displayed linear expression in B_t. Since B_t is elliptic, every non-scalar element of the L-algebra $L[B_t]$ has the same pair of fixed points at each real embedding. The chosen upper-half-plane fixed point is therefore carried by G from the companion Hecke fixed point to the B_t vertex. The accompanying exact certificate performs no floating-point comparison. □

5.3 From fixed matrices to endomorphisms of the explicit lattice

The preceding vertex certificate is an identity in the Hilbert modular matrix model. We now record explicitly why it is also an identity of endomorphisms of the polarized abelian variety constructed above in Section 4. This removes a possible ambiguity between a Möbius fixed-point computation and the actual rational Hodge structure.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathcal{O}_L)$$

and suppose that $z = (z_i) \in \mathbb{H}^6$ is fixed componentwise by A, so that

$$z_i = \frac{\sigma_i(a)z_i + \sigma_i(b)}{\sigma_i(c)z_i + \sigma_i(d)}.$$

Put $\lambda_i = \sigma_i(c)z_i + \sigma_i(d)$ and let m_λ denote the diagonal complex-linear map with multipliers λ_i.

Proposition 5.4 (Fixed matrices act on the explicit lattice)**.** *The map m_λ preserves Λ_z and hence induces an endomorphism of A_z. Relative to the $\mathcal{O}_L$-basis $(z, 1)$ of the rank-two $\mathcal{O}_L$-module defining Λ_z, its coordinate matrix is A^{T}. If $\det_L(A) = \delta \neq 0$, then*

$$[\Lambda_z : m_\lambda(\Lambda_z)] = |N_{L/\mathbb{Q}}(\delta)|,$$

so the induced isogeny has degree $|N_{L/\mathbb{Q}}(\delta)|$. In particular, $A_{c_{11}}$ induces a degree-43 endomorphism, while B_t induces an automorphism.

Proof. The fixed-point equation gives

$$\lambda_i z_i = \sigma_i(a)z_i + \sigma_i(b), \qquad \lambda_i = \sigma_i(c)z_i + \sigma_i(d).$$

Thus multiplication by λ sends the two $\mathcal{O}_L$-generators z and 1 to the $\mathcal{O}_L$-linear combinations encoded by the columns of A^{T}, proving lattice preservation and the matrix assertion. The index of the image of the rank-two $\mathcal{O}_L$-lattice under an L-linear matrix is the absolute norm of its L-determinant. For $A_{c_{11}}$ this is $N_{L/\mathbb{Q}}(\pi) = 43$, while $\det B_t = 1$. □

Now let $\gamma = \left(\begin{smallmatrix} r & s \\ u & v \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathcal{O}_L)$ and put $z' = \gamma z$. The standard analytic isomorphism

$$q_\gamma : A_z \longrightarrow A_{z'}, \qquad (w_i)_i \longmapsto \left(\frac{w_i}{\sigma_i(u)z_i + \sigma_i(v)}\right)_i$$

is an isomorphism of lattices. On the abstract $\mathcal{O}_L^2$ coordinates it is represented by $\gamma^{-\mathsf{T}}$. Since $\det \gamma = 1$, it preserves the determinant pairing and hence the trace polarization E_1.

Corollary 5.5 (The exact G-certificate transports the endomorphism)**.** *Under the polarized isomorphism q_G from the $k = 11$ companion fixed point to the normalized B_t-vertex, the endomorphism represented by $A_{c_{11}}^{\mathsf{T}}$ is carried to*

$$(GA_{c_{11}}G^{-1})^{\mathsf{T}} = (xI + yB_t)^{\mathsf{T}}.$$

Consequently the field generated by the corrected Hecke endomorphism is identified inside the actual endomorphism algebra of the certified vertex with the field generated by B_t^{T}. The K-operator $\Sigma_B = (I+2B_t^{14})^{\mathsf{T}}$ used below is therefore an endomorphism of the same polarized rational Hodge structure appearing in the exact Hecke certificate.

Proof. In lattice coordinates, conjugation by q_G sends A^{T} to

$$G^{-\mathsf{T}}A^{\mathsf{T}}G^{\mathsf{T}} = (GAG^{-1})^{\mathsf{T}}.$$

Apply Theorem 5.3. Preservation of E_1 follows from $G \in \mathrm{SL}_2(\mathcal{O}_L)$. □

5.4 The actual CM type at the vertex

There is a small orientation point here which is essential. McMullen's construction associates to the i-th real embedding a triangle $T_i \subset \mathbb{H}$; when the orientation of T_i agrees with T_1, the component map lands in $\mathbb{H}$, and when the orientations disagree it lands in $-\mathbb{H}$ [6, Thm. 6.3 and Eq. (6.1)]. Using McMullen's printed matrices A, B, C and their six real embeddings, the oriented vertex triples have relative orientation signs

$$(+, +, -, -, -, -)$$

for $k = 1, 5, 11, 13, 17, 19$. These are exactly the signs of $\sigma_k(t)$, where $t = 2\cos(\pi/21)$. Thus the fixed B-vertex contributes the eigenvalue ζ^k in the first two components and ζ^{-k} in the last four. This gives a geometric derivation of the CM-type signs below, independent of the normalizing matrix h_t alone. The signs are certified by rational isolating intervals for all six real roots of the minimal polynomial of t and exact rational interval evaluation of the three fixed points; no floating-point sign decision enters the proof.

Remark 5.6 (The caption of McMullen's Figure 5). McMullen's general construction is explicitly formulated in $(\pm\mathbb{H})^6$ and sends an oppositely oriented Galois triangle to $-\mathbb{H}$ [6, Thm. 6.3, Eq. (6.1)]. The caption of Figure 5 writes $\widetilde{f}_0 : \mathbb{H} \to \mathbb{H}^6$. For the printed matrix model, however, the direct certified orientation calculation above gives the mixed sign vector $(+, +, -, -, -, -)$. We therefore use the matrices and Theorem 6.3, rather than the caption alone, to determine the raw component of Y_L. Proposition S.7.1 then transfers that raw component to X_L by the integral matrix h_t.

Proposition 5.7 (CM type and Weil signature)**.** *The CM type of the vertex in Theorem 5.3 is*

$$\Phi = \{1, 5, 31, 29, 25, 23\} \subset (\mathbb{Z}/42\mathbb{Z})^\times.$$

For the quadratic subfield $K = \mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$, the restrictions of these six embeddings have residues

$$1, 2, 1, 2, 1, 2 \pmod 3.$$

Hence the K-action on $H^{1,0}$ has multiplicities $(3, 3)$: the certified vertex is an abelian sixfold of K-Weil type.

Proof. The sign statement above selects the exponents $1, 5, -11, -13, -17, -19$ modulo 42, which are exactly the displayed set. Restriction to $K = \mathbb{Q}(\zeta_3)$ is determined by the exponent modulo 3, giving three copies of each of the two embeddings of K. □

5.5 The Weil discriminant and Markman's theorem

Let $s = \sqrt{-3} \in K$, acting as Σ_B. With respect to E_1, define

$$H(v, w) = E_1(\Sigma_B v, w) + sE_1(v, w). \tag{5.1}$$

The identities $\Sigma_B^\dagger = -\Sigma_B$ and $\Sigma_B^2 = -3I$ imply $H(\Sigma_B v, w) = sH(v, w)$ and $H(v, \Sigma_B w) = \bar{s}H(v, w)$, so (5.1) is a K-Hermitian form for the polarized Weil structure (up to the conventional nonzero rational normalization, which does not change the determinant class in rank six).

Let $\eta \in L$ be the lower-left entry of the Mobius representative $S_B = I + 2B_t^{14}$. Exact reduction gives

$$\eta = 2(t^2 - 2)(2t^3 + t^2 - 7t - 4), \qquad N_{L/\mathbb{Q}}(\eta) = -1728.$$

The six vectors $f_j = (0, t^j)$, $0 \le j \le 5$, form a K-basis of L^2. Since $E_1(f_i, f_j) = 0$, the Hermitian matrix in this basis is

$$\left(-\operatorname{Tr}_{L/\mathbb{Q}}(\eta t^{i+j})\right)_{0 \le i,j \le 5}.$$

The basis assertion follows because $\eta \neq 0$: the vectors $f_j, \Sigma_B f_j$ span L^2 over $\mathbb{Q}$. Since the rank is six, the minus sign does not change the determinant. Therefore

$$\det H = \operatorname{disc}(L) N_{L/\mathbb{Q}}(\eta) = (3^3 7^5)(-1728) = -2^6 3^6 7^5. \tag{5.2}$$

Every identity here is exact in $\mathbb{Q}[t]/(f)$.

Proposition 5.8 (Weil discriminant -1)**.** *For the explicit polarization E_1, the K-Hermitian discriminant of the sixfold of Theorem 5.3 is the class of -1 in*

$$\mathbb{Q}^\times / N_{K/\mathbb{Q}}(K^\times).$$

Proof. Equation (5.2) represents the determinant class. Now

$$2^6 = N_{K/\mathbb{Q}}(8), \qquad 3^6 = N_{K/\mathbb{Q}}(27), \qquad 7 = N_{K/\mathbb{Q}}(2+\sqrt{-3}).$$

Therefore $2^6 3^6 7^5$ is a $K/\mathbb{Q}$ norm, leaving precisely the class -1. □

Markman's Section 1.1 defines a polarized abelian variety of Weil type as a triple (A, η, h) in which $h \in H^{1,1}(A, \mathbb{Q})$ is an ample class satisfying $\eta(k)^* h = N_{K/\mathbb{Q}}(k) h$; no principality hypothesis is imposed [4, Sec. 1.1]. In the present example the hypotheses match term by term: $\dim A_B = 6$; Proposition 5.7 gives multiplicities $(3, 3)$; E_1 is an ample integral, hence rational, $(1,1)$-class; Proposition 4.7 gives the required norm-similitude; and Proposition 5.8 gives discriminant class -1 for the associated K-Hermitian form. Markman's Theorem 1.5.1 therefore applies exactly and states that the Hodge–Weil classes are algebraic [4, Thm. 1.5.1]. The theorem does not exclude CM points.

Corollary 5.9 (Algebraicity of the exceptional Weil space)**.** *For the $k = 11$ vertex of Theorem 5.3, the two-dimensional rational Hodge–Weil space*

$$W_K(A) \subset H^6(A, \mathbb{Q}) \cap H^{3,3}(A)$$

is algebraic.

Proof. Proposition 5.7 gives Weil type for $K = \mathbb{Q}(\sqrt{-3})$, Proposition 5.8 gives discriminant -1, and Markman's theorem applies. □

5.6 All Hodge classes on the $k = 11$ CM sixfold

We finally determine the Hodge classes of this particular CM sixfold. This is stronger than merely identifying the exceptional Weil subspace.

Let $G = (\mathbb{Z}/42\mathbb{Z})^\times = \mathrm{Gal}(M/\mathbb{Q})$ and let e_a denote the one-dimensional complex eigenspace of $H^1(A, \mathbb{C})$ indexed by $a \in G$. For a subset $S \subset G$ put $e_S = \bigwedge_{a \in S} e_a$. If $|S| = 2p$, the standard CM-torus character criterion says that e_S is a Hodge line in $H^{2p}(A, \mathbb{C})$ precisely when

$$|gS \cap \Phi| = p \qquad \text{for every } g \in G. \tag{5.3}$$

To make the character calculation explicit, let $T = \mathrm{Res}_{M/\mathbb{Q}} \mathbf{G}_m$ and let μ be the Hodge cocharacter of the CM type Φ. The Mumford–Tate torus is the smallest $\mathbb{Q}$-subtorus of T through which all Galois conjugates $g\mu$ factor. If χ_S is the $T_{\mathbb{C}}$-character of e_S, then

$$\langle \chi_S, g\mu \rangle = |gS \cap \Phi|.$$

For $|S| = 2p$, the line e_S has Hodge type (p, p) exactly when this pairing equals the weight p for every Galois conjugate of the Hodge cocharacter. Hence

$$|gS \cap \Phi| = p \quad \text{for every } g \in G,$$

which is (5.3). Equivalently, after subtracting the central weight cocharacter, the character χ_S is trivial on the Hodge group. This is the elementary character form of the CM-torus description used in the general theory of Hodge classes on CM abelian varieties; compare [9].

Complex conjugation on G is $a \mapsto -a$. The six conjugate pairs are

$$\{1,41\},\{5,37\},\{11,31\},\{13,29\},\{17,25\},\{19,23\}.$$

Each corresponding degree-two Hodge line is a divisor line, and products of divisors correspond to unions of these pairs.

Lemma 5.10 (Fourier classification for the primitive CM type)**.** *Let*

$$\Phi = \{1,5,23,25,29,31\} \subset G$$

and let $S \subset G$ have cardinality $2p$. Then

$$|gS \cap \Phi| = p \quad \textit{for every } g \in G$$

if and only if either S is a union of p antipodal pairs $\{a,-a\}$, or $p=3$ and

$$S = \{a \in G : a \equiv 1 \pmod 3\} \quad \textit{or} \quad S = \{a \in G : a \equiv 2 \pmod 3\}.$$

Consequently the numbers of solutions for $p = 1,\dots,6$ are

$$\binom{6}{p} + 2\delta_{p,3} = 6,15,22,15,6,1.$$

The same classification holds for every Galois translate of Φ.

Proof. Use the Chinese-remainder coordinates

$$G \simeq C_2 \times C_6, \qquad a \bmod 3 = (-1)^e, \qquad a \bmod 7 = 3^n,$$

where $e \in \{0,1\}$ and $n \in \mathbb{Z}/6\mathbb{Z}$. In these coordinates complex conjugation $a \mapsto -a$ is translation by $(1,3)$. Put $\varepsilon = 2\mathbf{1}_\Phi - 1$. With columns indexed by $n = 0,\dots,5$, direct CRT reduction gives

$$\big(\varepsilon(0,n)\big)_n = (1,1,-1,-1,1,-1), \qquad \big(\varepsilon(1,n)\big)_n = (1,-1,1,-1,-1,1).$$

In particular $\varepsilon(x+(1,3)) = -\varepsilon(x)$.

Let ζ_6 be a primitive sixth root of unity and use the Fourier convention

$$\widehat{f}(r,s) = \sum_{e=0}^{1}\sum_{n=0}^{5} f(e,n)(-1)^{re}\zeta_6^{-sn}, \qquad r \in \mathbb{Z}/2\mathbb{Z}, \quad s \in \mathbb{Z}/6\mathbb{Z}.$$

The character $\chi_{r,s}$ is odd under complex conjugation exactly when $r+s$ is odd. Since ε is odd, its even Fourier coefficients vanish. For the six odd characters, summing the two displayed rows and using $\zeta_6^2 - \zeta_6 + 1 = 0$ gives the exact table

(r,s)	$(0,1)$	$(0,3)$	$(0,5)$	$(1,0)$	$(1,2)$	$(1,4)$
$\widehat{\varepsilon}(r,s)$	4	4	4	0	$-4(2\zeta_6-1)$	$4(2\zeta_6-1)$.

Thus the only odd character missing from the Fourier support of ε is

$$\eta = \chi_{1,0}, \qquad \eta(e,n) = (-1)^e,$$

the quadratic character modulo 3.

Let $F = \mathbf{1}_S$ and define the correlation

$$C_F(g) = \sum_{x \in G} F(x)\varepsilon(g^{-1}x) = 2|S \cap g\Phi| - |S|.$$

The Hodge condition is equivalent to $C_F = 0$, since g ranges through all of G. Character orthogonality, with the convention above, gives

$$\widehat{C}_F(\chi) = \widehat{F}(\chi)\widehat{\varepsilon}(\chi^{-1}).$$

The odd support in the table is invariant under inversion. Therefore every odd Fourier coefficient of F vanishes except possibly the η-coefficient. Equivalently the odd part of F has the form

$$F^-(x) := \frac{F(x) - F(-x)}{2} = c\eta(x)$$

for some real number c. Because F is binary-valued, $F(x) - F(-x) \in \{-1, 0, 1\}$. If $c = 0$, F is constant on each antipodal pair, so S is a union of such pairs. If $c \neq 0$, the difference is nonzero and has the same absolute value on all six pairs; hence $c = \pm\frac{1}{2}$ and each pair contributes exactly one element. Solving for F gives $F = (1 + \eta)/2$ or $F = (1 - \eta)/2$, the two residue fibers modulo 3. They both have six elements, so they occur only when $p = 3$.

There are $\binom{6}{p}$ unions of p antipodal pairs. Finally, if $\Phi' = u\Phi$, then $\widehat{\varepsilon}_{\Phi'}(\chi) = \chi(u)^{-1}\widehat{\varepsilon}_{\Phi}(\chi)$, so its Fourier support is unchanged. This proves the translated statement as well. □

The same finite Galois calculation shows that the stabilizer of Φ in G is trivial. Hence this CM type is primitive: it is not pulled back from a CM type on any proper CM subfield of M. In particular, the example is not produced by the obvious induction of a lower-dimensional CM type.

Theorem 5.11 (Hodge ring of the certified CM sixfold)**.** *For the CM type*

$$\Phi = \{1, 5, 23, 25, 29, 31\},$$

the numbers of complex Hodge eigenlines in codimensions $p = 1, \ldots, 6$ are

$$6,\ 15,\ 22,\ 15,\ 6,\ 1.$$

In every codimension except $p = 3$, all of these lines are products of divisor lines. In codimension 3, exactly 20 of the 22 lines are products of three divisor lines; the two remaining lines are indexed by

$$S_+ = \{1, 13, 19, 25, 31, 37\}, \qquad S_- = \{5, 11, 17, 23, 29, 41\}.$$

These are precisely the two complex eigenspaces of $\bigwedge^6_K H^1(A, \mathbb{Q})$; their rational span is $W_K(A)$. Consequently the full rational Hodge algebra of A is generated by divisor classes together with $W_K(A)$.

Proof. Lemma 5.10 classifies every binary solution of criterion (5.3), giving the displayed Hodge counts. The six degree-two solutions are precisely the antipodal pairs. Their exterior products are nonzero exactly when the chosen pairs are distinct, so they give $\binom{6}{p}$, namely $6, 15, 20, 15, 6, 1$, divisor-product eigenlines. The only two additional lines occur in the middle degree. The two sets displayed above are exactly the fibers $a \equiv 1 \pmod 3$ and $a \equiv 2 \pmod 3$, hence the two eigenspaces obtained after complexifying $\bigwedge^6_K H^1$. This identifies their rational Galois-stable span with the two-dimensional Weil space. The registered exact subset enumerations independently verify the

classification and counts, but are not the sole explanation. Although the eigenline description is made after complexification, the divisor eigenlines are permuted by G and their full span is the complexification of the rational divisor space; likewise S_+ and S_- form one Galois-stable pair. Hence the resulting spanning statement descends from complex eigenspaces to the rational Hodge algebra. □

Corollary 5.12 (Hodge conjecture for the certified sixfold)**.** *The Hodge conjecture holds in every codimension for the specific polarized CM abelian sixfold represented by the $k = 11$ normalized B-vertex.*

Proof. Divisor classes are algebraic by the Lefschetz $(1,1)$ theorem [1, §2, Remarks (iii), p. 49], hence so are all their products. Corollary 5.9 gives algebraicity of the only two Hodge eigenlines not contained in the divisor algebra. Theorem 5.11 exhausts all Hodge classes in all degrees. □

5.7 The A and C realizations and the fourth vertex

Theorem 5.13 (A second exact Hecke vertex)**.** *Let A be McMullen's order-14 lift of the projective order-7 quadrilateral vertex generator in the index-two $(7, 21, 21, 21)$ model, and put*

$$A_t = h_t A h_t^{-1}, \qquad h_t = \mathrm{diag}(t, 1).$$

For every one of the four $d = 7$ arithmetic signature candidates

$$k \in \{2, 14, 23, 35\}$$

there are explicit $x_k, y_k \in L$ such that

$$T_k = x_k I + y_k A_t \in M_2(\mathcal{O}_L), \qquad \mathrm{tr}(T_k) = c_k, \qquad \det(T_k) = \pi.$$

Thus T_k is an integral degree-43 Hecke representative and has exactly the same componentwise fixed points as A_t. Consequently all four corrected $d = 7$ trace classes admit a Hecke representative fixing the normalized A-vertex of McMullen's curve.

The CM type of the resulting A-vertex rational Hodge structure is

$$\Phi_A = \{1, 5, 31, 29, 17, 19\}.$$

Its restriction to $\mathbb{Q}(\sqrt{-7})$ has multiplicities $(2, 4)$, so the A-vertex is not *of $\mathbb{Q}(\sqrt{-7})$-Weil type despite the arithmetic $d = 7$ coefficient-sign filter. Its restriction to $K = \mathbb{Q}(\sqrt{-3})$, however, has multiplicities $(3, 3)$.*

Proof. McMullen's matrix A has trace

$$\mathrm{tr}(A) = t^3 - 3t = \zeta_{42}^3 + \zeta_{42}^{-3}.$$

Hence $L[A_t] = M$. Writing $\lambda_A = \zeta_{42}^3$, each α_k has a unique expansion

$$\alpha_k = x_k + y_k \lambda_A.$$

For the two conjugate pairs these coefficients are

$$x_2 = \frac{3t^5 - t^4 - 19t^3 + 5t^2 + 29t - 10}{7},$$
$$y_2 = -\frac{4t^5 + t^4 - 30t^3 - 5t^2 + 48t - 4}{7},$$
$$x_{14} = \frac{3t^5 - t^4 - 26t^3 - 2t^2 + 43t + 4}{7},$$
$$y_{14} = \frac{(t-1)(3t^4 + 9t^3 - 3t^2 - 26t - 18)}{7},$$

and

$$(x_{23}, y_{23}) = -(x_2, y_2), \qquad (x_{35}, y_{35}) = -(x_{14}, y_{14}).$$

Although the displayed coefficients have denominator 7 in the power basis, exact reduction in $L = \mathbb{Q}[t]/(f)$ shows that all four matrices $T_k = x_k I + y_k A_t$ have entries in $\mathcal{O}_L$ and satisfy the displayed trace and determinant identities. Since $y_k \neq 0$, the fixed-point equation of T_k is a nonzero scalar multiple of that of A_t, so their fixed points agree at every real embedding. Because $N_{L/\mathbb{Q}}(\pi) = 43$ is prime, an integral matrix of determinant π has the required local Hecke elementary-divisor type.

For the CM type, the normalized A_t fixed point lies in $\mathbb{H}$ at every real place. If $c_{A,t}$ denotes the lower-left entry of A_t, the eigenvalue on the $H^{1,0}$ fixed line has imaginary part $c_{A,t}\,\mathrm{Im}(z)$. Exact interval evaluation at the six real embeddings, compared with the signs of $\sin(k\pi/7)$ for the eigenvalues $\zeta_{42}^{\pm 3k}$, selects the extensions

$$1, 5, -11, -13, 17, 19 \pmod{42},$$

which gives the displayed Φ_A. Reduction modulo 7 gives two quadratic-residue and four nonresidue restrictions, while reduction modulo 3 gives three of each embedding. This proves the two signature assertions. □

Theorem 5.14 (Hodge conjecture at the normalized A-vertex)**.** *For the normalized A-vertex, every rational Hodge class is generated by divisor classes. In particular the Hodge conjecture holds for the associated polarized abelian sixfold in every codimension.*

Moreover, for the $K = \mathbb{Q}(\sqrt{-3})$ Weil structure its Hermitian discriminant is the class of -1.

Proof. The element $\sqrt{-3} = 1 + 2\zeta_{42}^{14}$ has a unique expression $x + y\zeta_{42}^3$ over L. Replacing ζ_{42}^3 by A_t gives an integral Mobius representative S_A with $S_A^2 = -3I$. Its actual homology action is $\Sigma_A = S_A^{\mathsf{T}}$. For the trace polarization E_1, the corresponding Hermitian form

$$H_A(v, w) = E_1(\Sigma_A v, w) + \sqrt{-3}\, E_1(v, w)$$

has Gram matrix $(-\operatorname{Tr}_{L/\mathbb{Q}}((S_A)_{21} t^{i+j}))$ in the K-basis $(0, t^j)$, and exact determinant

$$\det H_A = -2^6 3^6 7^5\, 127.$$

Here $2^6 = N(8)$, $3^6 = N(27)$, $7 = N(2 + \sqrt{-3})$, and

$$127 = N(10 + 3\sqrt{-3}),$$

so the determinant class is -1.

It remains to determine all Hodge classes. Applying the CM-character criterion to

$$\Phi_A = \{1, 5, 31, 29, 17, 19\}$$

gives respectively

$$12,\ 51,\ 88,\ 51,\ 12,\ 1$$

complex Hodge eigenlines in codimensions $1, \ldots, 6$. There are exactly twelve degree-two Hodge eigenlines, indexed by

$$\{1, 13\},\ \{1, 41\},\ \{5, 23\},\ \{5, 37\},\ \{11, 17\},\ \{11, 31\},$$
$$\{13, 29\},\ \{17, 25\},\ \{19, 23\},\ \{19, 37\},\ \{25, 31\},\ \{29, 41\}.$$

By the Lefschetz $(1,1)$ theorem their rational Galois-stable span consists of divisor classes. In particular the two $K=\mathbb{Q}(\sqrt{-3})$ Weil eigenlines are already divisor products; for example

$$\{a : a \equiv 1 \pmod 3\} = \{1,13\} \sqcup \{19,37\} \sqcup \{25,31\},$$

with the conjugate line decomposing as

$$\{5,23\} \sqcup \{11,17\} \sqcup \{29,41\}.$$

Exhaustive exact enumeration of disjoint products of the twelve degree-two eigenlines gives

$$12,\ 51,\ 88,\ 51,\ 12,\ 1$$

distinct eigenlines in the corresponding codimensions, exactly exhausting the preceding Hodge counts. Thus the complexified Hodge algebra is generated in degree two, and Galois stability descends this statement to the rational Hodge algebra. All such generators are algebraic divisors, proving the Hodge conjecture in every codimension. □

Proposition 5.15 (The normalized C-vertex and saturation of the 43-packet)**.** *Let $C_t = h_t C h_t^{-1}$ be the normalized second order-42 generator. Then*

$$\operatorname{tr}(C_t) = t, \qquad \det(C_t) = 1, \qquad \mathcal{O}_L[C_t] = \mathcal{O}_M.$$

For every $k = 0, \dots, 41$, substitution $\zeta_{42} \mapsto C_t$ gives

$$T_{C,k} = \alpha_k(C_t) \in M_2(\mathcal{O}_L), \qquad \operatorname{tr}(T_{C,k}) = c_k, \qquad \det(T_{C,k}) = \pi.$$

Thus every corrected $\ell = 43$ trace class has a representative fixing the normalized C-vertex. The identical statement holds at B_t.

The CM type at C is

$$\Phi_C = \{1,5,11,13,17,19\}.$$

It has $K = \mathbb{Q}(\sqrt{-3})$-signature $(3,3)$ and K-Hermitian discriminant -1. The Hodge eigenline counts are

$$6,\ 15,\ 22,\ 15,\ 6,\ 1,$$

and the only two non-divisor lines are the two complex lines of $W_K(A_C)$. Consequently the Hodge conjecture holds for the polarized C-vertex in every codimension.

Proof. The relative discriminant of $X^2 - tX + 1$ is $t^2 - 4$, and exact norm computation gives $N_{L/\mathbb{Q}}(t^2-4) = 1$. Hence the order generated by either B_t or C_t is the maximal order $\mathcal{O}_M$. Evaluating the integral element $\alpha_k = \zeta_{42}^k(1 + \zeta_{42} + \zeta_{42}^4)$ at either matrix therefore gives an integral endomorphism. Its L-trace and determinant are c_k and $N_{M/L}(\alpha_k) = \pi$. Since $N_{L/\mathbb{Q}}(\pi) = 43$ is prime, the matrix has the required local Hecke elementary-divisor type.

Exact interval evaluation shows that the lower-left entry of C_t is positive at all six real embeddings, so the upper fixed lines select the exponents $1,5,11,13,17,19$. Put $S_C = I + 2C_t^{14}$. Then the actual homology action $\Sigma_C = S_C^{\mathsf{T}}$ satisfies $\Sigma_C^2 = -3I$, $\Sigma_C^\dagger = -\Sigma_C$. In the K-basis $(0, t^j)$ the Hermitian Gram matrix is $(-\operatorname{Tr}_{L/\mathbb{Q}}((S_C)_{21} t^{i+j}))$, and its determinant is

$$-2^6 3^6 7^5\, 127.$$

The factors outside the sign are norms because $7 = N(2+\sqrt{-3})$ and $127 = N(10+3\sqrt{-3})$, leaving discriminant class -1. The exact CM-character enumeration then gives the displayed Hodge counts and exactly two exceptional middle-degree lines, namely the two K-Weil eigenspaces. Markman's theorem applies. □

Proposition 5.16 (The B- and C-vertices are distinct holomorphic classes)**.** *The normalized B- and C-vertices are not $\mathrm{SL}_2(\mathcal{O}_L)$-equivalent, although B_t and C_t are in the same unoriented $\mathrm{GL}_2(\mathcal{O}_L)$ similarity class.*

Proof. The maximal order $\mathcal{O}_M$ has class number one and $h(L) = 1$, so the maximal-order module classification gives a single $\mathrm{GL}_2(\mathcal{O}_L)$ similarity class. Let P conjugate B_t to C_t. At the real embeddings $k = 1, 5$ the chosen upper fixed lines carry the same eigenvalue ζ_{42}^{k}, while at $k = 11, 13, 17, 19$ the B_t upper line carries ζ_{42}^{-k} and the C_t upper line carries ζ_{42}^{k}. Thus

$$\operatorname{sgn}(\det P) = (+, +, -, -, -, -).$$

Every other integral conjugator differs by a centralizer unit in $\mathcal{O}_M^{\times}$, whose determinant is a relative unit norm and is totally positive. Hence this sign vector cannot be changed. No integral conjugator is totally positive, and in particular none lies in $\mathrm{SL}_2(\mathcal{O}_L)$. □

Proposition 5.17 (The fourth quadrilateral vertex is the B-class)**.** *Let D be McMullen's fourth printed generator for the symmetric quadrilateral $Q(7, 21, 21, 21)$ and put $D_t = h_t D h_t^{-1}$. Then*

$$\operatorname{tr}(D_t) = t, \qquad \det(D_t) = 1, \qquad \mathcal{O}_L[D_t] = \mathcal{O}_M,$$

and D_t is $\mathrm{SL}_2(\mathcal{O}_L)$-conjugate to B_t. Consequently the four elliptic vertices A, B, C, D of McMullen's quadrilateral determine exactly three holomorphic classes in the normalized Hilbert modular quotient: A, $B = D$, and C.

Proof. McMullen's explicit matrix for D has trace t and determinant one. Since the relative discriminant of $X^2 - tX + 1$ is $t^2 - 4$ and $N_{L/\mathbb{Q}}(t^2 - 4) = 1$, both $\mathcal{O}_L[B_t]$ and $\mathcal{O}_L[D_t]$ equal the maximal order $\mathcal{O}_M$. The maximal-order integral similarity theorem therefore puts B_t and D_t in one $\mathrm{GL}_2(\mathcal{O}_L)$ class.

Let $B = \left(\begin{smallmatrix} 0 & -1 \\ 1 & t \end{smallmatrix}\right)$ be the companion matrix for $X^2 - tX + 1$. The rational matrix $P_0 = (e_1, D_t e_1)$ satisfies $P_0 B = D_t P_0$, and exact rational-interval evaluation gives

$$\operatorname{sgn}(\det P_0) = (+, +, -, -, -, -).$$

The same sign vector occurs for $h_t = \operatorname{diag}(t, 1)$, which satisfies $h_t B = B_t h_t$. Therefore $P_0 h_t^{-1}$ conjugates B_t to D_t and has totally positive determinant. Any two conjugators between B_t and D_t differ through the centralizer $M^{\times}$, whose determinant is a relative norm and hence positive at every real place. Thus an integral conjugator supplied by the maximal-order classification lies in the same positive orientation class; Corollary S.11.4 adjusts it through an integral centralizer unit to determinant one. Hence B_t and D_t are $\mathrm{SL}_2(\mathcal{O}_L)$-conjugate. □

5.8 Comparison of the underlying isogeny classes

Proposition 5.18 (Comparison with known CM factors)**.** *Let J_n be the Jacobian of the smooth projective model of $y^2 = x^n - 1$, and let r_n be the endomorphism induced by $(x, y) \mapsto (\zeta_n x, y)$. Put $X_{21} = \operatorname{im}(\Phi_3(r_{21})\Phi_7(r_{21}) : J_{21} \to J_{21})$. Forgetting the chosen polarizations and fixed RM markings, the three fibers of Theorem 1.2 satisfy*

$$A_A \sim_{\mathbf{C}} J_7^2, \qquad A_B \sim_{\mathbf{C}} A_C \sim_{\mathbf{C}} X_{21}.$$

Here X_{21} is a simple abelian sixfold. Thus the three marked quotient classes belong to exactly two underlying complex isogeny classes.

Proof. For odd n, the differentials $x^{j-1}dx/y$, $1 \le j \le (n-1)/2$, have r_n-characters ζ_n^j. We use pushforward on divisor classes; equivariance of the Abel–Jacobi map identifies these with the Jacobian tangent eigenvalues. The polynomial $\Phi_3\Phi_7$ kills precisely the nonprimitive characters of J_{21}, so X_{21} has dimension six and CM type $S = \{1, 2, 4, 5, 8, 10\}$ for $\xi = \zeta_{21}$. This is the known primitive factor in [2, Eq. (4.2), Lem. 4.2, Prop. 4.1 and Sec. 5.3]. Its type has trivial stabilizer in $(\mathbf{Z}/21\mathbf{Z})^\times$, hence is primitive and gives a simple variety [8, Prop. 3.13].

Since $z = \zeta_{42} = -\xi^{11}$, the action $z \mapsto -r_{21}^{11}|_{X_{21}}$ has type $\{21 + 22j \bmod 42 : j \in S\} = \Phi_B$. Precomposing this action by $z \mapsto z^{19}$ gives $19\Phi_B = \Phi_C$. This last automorphism acts nontrivially on L; it does not identify the two prescribed RM markings.

For A, on J_7^2 define commuting endomorphisms

$$R = \operatorname{diag}(r_7, r_7), \quad U = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad T = -R^2U^2 = \begin{pmatrix} r_7^2 & -r_7^2 \\ r_7^2 & 0 \end{pmatrix}.$$

The identities $R^7 = I$ and $U^2 + U + I = 0$ give $T^6 = R^5$, $T^{21} = -I$ and $T^{42} = I$. Its tangent eigenvalues have z-exponents $21 + 12j + 28b \pmod{42}$ for $j = 1, 2, 3$ and $b = 1, 2$. Their set is $\{1, 5, 17, 19, 29, 31\} = \Phi_A$. All are primitive, so $\Phi_{42}(T) = 0$ and $z \mapsto T$ gives a faithful M-action. No compatibility with the product polarization is asserted.

Each compared sixfold has rational homology of rank one over the degree-twelve field M. Equal specified CM types identify their rational complex structures; the corresponding full lattices in M are commensurable. Multiplication by a suitable positive integer therefore gives an M-equivariant complex isogeny, which is algebraic. This is the CM classification up to isogeny [8, Prop. 3.12], and it does not require the varieties to be simple. Finally, J_7^2 is not simple, so its isogeny class differs from that of X_{21}. □

6 A common principal lattice and family

The construction is a change of integral lattice, not a declaration that the original trace polarization is principal. The subgroup at 7 is nonfree; requiring a common free line would incorrectly rule it out. The special-fiber K-actions need not extend to a single global K-action.

Proposition 6.1 (A common marked principal lattice at the three vertices)**.** *Use the common marking $\Lambda = \mathcal{O}_L^2$, with $(a, b) \mapsto az_X + b$ at $X = A, B, C$. There is one self-dual overlattice $\Lambda \subset \Lambda^{\mathrm{common}} \subset \Lambda^\#$ of index 453789 stable under $\mathcal{O}_L$ and all three specialized $\mathcal{O}_K$-actions $\Sigma_X = S_X^{\mathsf{T}}$. With the respective complex structures it gives three principally polarized isogenous special fibers with unchanged rational CM Hodge structures.*

Proof. Put $R = \mathcal{O}_L/\mathfrak{p}_7^5 = \mathbb{F}_7[u]/(u^5)$, where $u = t - 1$. The actual homology residues are

$$\Sigma_A \equiv \begin{pmatrix} 1 & 3 \\ 1 & 6 \end{pmatrix}, \qquad \Sigma_B \equiv \begin{pmatrix} 6 & 2 \\ 5 & 1 \end{pmatrix}, \qquad \Sigma_C \equiv \begin{pmatrix} 5 & 5 \\ 0 & 2 \end{pmatrix} \pmod{u}.$$

They all preserve the line of $v = (3, 1)$. Define

$$H_7 = u^2Rv + u^3R^2.$$

Taking $w = (1, 0)$, the vectors v, w are an R-basis and $H_7 = u^2Rv \oplus u^3Rw$ has $\mathbb{F}_7$-basis

$$u^2v,\ u^3v,\ u^4v,\ u^3w,\ u^4w.$$

Thus $|H_7| = 7^5$. For each X, write $\Sigma_X v = \lambda_X v + uq_X$; this immediately proves $\Sigma_X H_7 \subset H_7$. The determinant of any two elements of H_7 vanishes modulo u^5.

At $\mathfrak{p}_3$, all three Σ_X vanish. Choose the same first-coordinate $\mathbb{F}_{27}$-line H_3, and put $H = H_3 \oplus H_7$. To verify isotropy for the actual discriminant pairing, let $d = f'(t)$. Multiplication by d identifies $\Lambda^\#/\Lambda$ with $(\mathcal{O}_L/(d))^2$. Representatives $x, y \in \Lambda$ of elements of H have $\det(x, y) \in (d)$, by the two local determinant calculations. Hence

$$E_1(d^{-1}x, d^{-1}y) = -\operatorname{Tr}_{L/\mathbb{Q}}\left(\frac{\det(x,y)}{d^2}\right) \in \mathbb{Z},$$

since the trace argument lies in the codifferent $d^{-1}\mathcal{O}_L$. As $|H| = 3^3 7^5 = \sqrt{|\Lambda^\#/\Lambda|}$, H is maximal isotropic. Exact power-basis reduction verifies that all three $U_X = (\Sigma_X - I)/2$ are integral. They preserve $\Lambda^\#$ as L-linear integral matrices; since 2 is invertible on the discriminant module, they preserve H. Thus its inverse image is simultaneously $\mathcal{O}_L$- and $\mathcal{O}_K$-stable and self-dual. The unchanged positive Riemann form supplies the asserted principal polarizations. □

Remark 6.2 (Free lines and the scope of the common construction). The common H_7 above is nonfree. Exhaustion of $\mathbb{P}^1(\mathbb{F}_7)$ shows that each pair of the three homology operators has only the common residue line $(3, 1)$. Its free lifts modulo u^2 have the form $(3 + au, 1)$, with respective unique coefficients

$$a_A = 3, \qquad a_B = 6, \qquad a_C = 2.$$

Consequently no pair admits a common invariant *free rank-one summand* on R^2. This restricted obstruction does not exclude the common nonfree maximal isotropic subgroup. Proposition 6.1 concerns three specialized K-actions on one marked lattice. The global construction below gives a family with RM and these three special fibers, without asserting a single global K-action or a realization of Mostaed's literal principally polarized locus.

Proposition 6.3 (Ambient and curve stabilizers of the common level)**.** *Let $\Gamma = \mathrm{SL}_2(\mathcal{O}_L)$, let $j : \Delta_0 \to \Gamma$ be the normalized integral representation of McMullen's quadrilateral group, and let H be the subgroup in Proposition 6.1, with $H_3 = \mathbb{F}_{27}(1, 0)$. For the transition representation $\rho(g) = g^{-\mathsf{T}}$ put*

$$\Gamma_H = \{g \in \Gamma : \rho(g)H = H\}, \qquad \Delta_H = j^{-1}(\Gamma_H).$$

Then

$$[\Gamma : \Gamma_H] = 224, \qquad [\Delta_0 : \Delta_H] = [\bar{\Delta}_0 : \bar{\Delta}_H] = 28.$$

The exact ideal congruence level of Γ_H is $\mathfrak{p}_3\mathfrak{p}_7$: it contains $\Gamma(\mathfrak{p}_3\mathfrak{p}_7)$, and $\Gamma(J) \subset \Gamma_H$ implies $J \subset \mathfrak{p}_3\mathfrak{p}_7$. The number 28 is the covering degree over the explicit parametrizing orbifold $\mathbb{H}/\bar{\Delta}_0$, before an auxiliary neat refinement.

Proof. With the marking $(a, b) \mapsto az + b$, the isomorphism from the fiber at z to the fiber at gz is $w \mapsto w/(cz + d)$. Its homology matrix is $g^{-\mathsf{T}}$. This transition action is a representation; it differs from the fixed-point endomorphism matrix g^{T}.

For any residue line $\ell \subset \mathbb{F}_7^2$, write $H(\ell) = u^2 Rv + u^3 R^2$, where v lifts ℓ. An invertible matrix T over R preserves $H(\ell)$ precisely when its reduction preserves ℓ. Necessity follows by projecting to u^2R^2/u^3R^2. Conversely, $Tv = \lambda v + uq$ gives $TH(\ell) \subset H(\ell)$, and finiteness gives equality. Thus Γ_H is the inverse image of two line stabilizers in $\mathrm{SL}_2(\mathbb{F}_{27}) \times \mathrm{SL}_2(\mathbb{F}_7)$. Reduction is surjective: the finite groups are generated by elementary matrices, whose entries lift by the Chinese remainder theorem. Hence the ambient index is $(27 + 1)(7 + 1) = 224$.

This description proves containment of $\Gamma(\mathfrak{p}_3\mathfrak{p}_7)$. If J is not contained in one of these primes, an elementary unipotent with entry in J can move the corresponding residue line. At 7 use the integral determinant-one basis with columns $(3,1),(-1,0)$. Applying ρ^{-1} gives an element of $\Gamma(J)$ outside Γ_H, proving the asserted exactness of the level.

McMullen's generators A, B, C, D generate Δ_0 and satisfy $ABCD = I$ [6, Sec. 4], so A, B, C suffice. Exact reduction of $\rho(j(A)), \rho(j(B)), \rho(j(C))$ on the 224-element set $\mathbb{P}^1(\mathbb{F}_{27}) \times \mathbb{P}^1(\mathbb{F}_7)$ fixes $(3,1)$ in the second factor. Exhaustive closure of $(\mathbb{F}_{27}(1,0), \mathbb{F}_7(3,1))$ is exactly

$$\mathbb{P}^1(\mathbb{F}_{27}) \times \{\mathbb{F}_7(3,1)\}.$$

The two separate implementations in Supplement, Section S.13 verify the permutations and this entire finite orbit. Forward closure suffices because inverses are positive powers of finite permutations. Orbit–stabilizer therefore gives 28, without a surjectivity assumption for the curve group. Both Δ_0 and Δ_H contain $-I$, so projectivization preserves this index. □

Theorem 6.4 (A connected principally polarized RM family)**.** *The normalized modular embedding admits a connected finite cover C' of its parametrizing orbifold and algebraic RM abelian schemes $\mathcal{A}, \mathcal{B}$ of relative dimension six over C', with an $\mathcal{O}_L$-linear isogeny*

$$q : \mathcal{A} \longrightarrow \mathcal{B}, \qquad \deg(q) = 453789, \qquad q^*\lambda_{\mathcal{B}} = \lambda_{\mathcal{A}}.$$

Here $\lambda_{\mathcal{A}}$ is the original trace polarization and $\lambda_{\mathcal{B}}$ is principal. The same connected cover contains lifts of all three specified normalized vertices whose $\mathcal{B}$-fibers are the common-lattice models of Proposition 6.1. Their specialized $\mathcal{O}_K$-actions and rational CM Hodge structures are preserved, so the Hodge conjecture holds for these three fibers.

The moduli image lies in the principally polarized Hilbert–Blumenthal component determined by the self-dual module $\Lambda_{\mathrm{common}}$. No identification with Mostaed's named curve or global K-action is included. The degree of a chosen auxiliary fine-level cover is not asserted to be 28*.*

Proof. Choose a neat finite-index subgroup $\Gamma_{\mathrm{aux}} \subset \Gamma_H$; it exists by [7, Thm. 3.6], since Γ_H is arithmetic in $\mathrm{Res}_{L/\mathbb{Q}}\mathrm{SL}_2$. Set $\Delta' = j^{-1}(\Gamma_{\mathrm{aux}})$ and $C' = \mathbb{H}/\bar{\Delta}'$. The group $\bar{\Delta}'$ is torsion-free and finite index in the cocompact quadrilateral group, so C' is a connected compact Riemann surface, hence a smooth projective algebraic curve. Its map to the parametrizing orbifold is finite; the coarse map may ramify at elliptic points. Since the neat group excludes $-I$, its geometric degree is measured using projective group indices.

Over $\mathbb{H}$ use the fixed normalized embedding F_t and the constant lattices $\Lambda = \mathcal{O}_L^2$ and $\Lambda' = \Lambda_{\mathrm{common}}$. The transition action ρj descends both to integral local systems. Their homology Hodge filtration F^0 is the kernel of the holomorphic map

$$(a,b) \longmapsto \big(\sigma_i(a)F_{t,i}(z) + \sigma_i(b)\big)_{i=1}^{6}.$$

It has rank six and Hodge types $\{(-1,0),(0,-1)\}$; Griffiths transversality is automatic for these types. The form E_1 is flat, integral and positive by the Riemann-form identity already proved, and is unimodular on Λ'. Thus both are polarizable integral variations of Hodge structure. The discriminant local system $\Lambda^{\#}/\Lambda$ has invariant subgroup H, and its inverse-image lattice sheaf is exactly Λ'. This identifies the monodromy descent of the common level.

Milne's equivalence for polarizable integral variations of these homology types over smooth complex algebraic varieties [7, Thm. 7.13] algebraizes both families; the proof explicitly includes nonprincipal polarizations. Full faithfulness algebraizes the lattice inclusion, the $\mathcal{O}_L$-actions and

the polarizations. The inclusion has index 453789, and both forms are E_1, giving the degree and pullback identity. Real multiplication is self-adjoint and $\lambda_{\mathcal{B}}$ is principal.

Finally take the original three universal-cover parameters z_A, z_B, z_C. Their images lie on the same connected curve $\mathbb{H}/\bar{\Delta}'$. The constant marking Λ' makes these fibers precisely the three models in Proposition 6.1. No choice of components of a fiber product is involved. The specialized K-actions need not be one global Hodge endomorphism. The RM family defines the stated moduli map, and additional level gives a fine moduli interpretation. □

Proposition 6.5 (The surviving order in the selected CM field)**.** *For each of the three common principal fibers B_X, let $\iota_X : M \hookrightarrow \mathrm{End}^0(B_X)$ be the selected CM field embedding. Then*

$$\mathrm{End}(B_X) \cap \iota_X(M) = \iota_X(R_K), \qquad R_K = \mathcal{O}_L[\mathcal{O}_K] = \mathcal{O}_L + \mathfrak{p}_3\mathcal{O}_M.$$

This order has index 27 in $\mathcal{O}_M$ and conductor $\mathfrak{p}_3\mathcal{O}_M$. The retained powers of the actual vertex automorphism are $0, 7$ modulo 14 at A, and $0, 7, 14, 21, 28, 35$ modulo 42 at B, C. The intersection assertion does not identify the full endomorphism ring at the imprimitive A-fiber, or the generic endomorphism ring of the family.

Proof. Put $z = \zeta_{42}$, so $\mathcal{O}_M = \mathcal{O}_L + \mathcal{O}_L z$. Exact arithmetic gives

$$z^{14} = a + bz, \quad \begin{cases} a = -t^5 + 4t^3 - 3t - 1, \\ b = -t^5 + t^4 + 6t^3 - 2t^2 - 8t - 2. \end{cases}$$

Both implementations verify $N_{L/\mathbb{Q}}(b) = -27$ and $b \equiv 0 \pmod{\mathfrak{p}_3}$. Thus $(b) = \mathfrak{p}_3$, proving the order formula and index. If $x = a_0 + b_0 z$ and both x, xz lie in R_K, their z coefficients force $a_0, b_0 \in \mathfrak{p}_3$; the reverse inclusion is immediate. This proves the conductor assertion.

The intersection order R' contains R_K by the common-lattice theorem and lies in $\mathcal{O}_M$, since lattice endomorphisms are integral over $\mathbb{Z}$. The quotient $\mathcal{O}_M/R_K \simeq \mathcal{O}_L/\mathfrak{p}_3$ is a simple $\mathcal{O}_L$-module. Consequently R' is either R_K or $\mathcal{O}_M$. The latter would preserve the full vertex automorphism, contradicted by the $\mathfrak{p}_3$ obstruction proved above for all three vertices. Hence $R' = R_K$. Exhausting the 14 or 42 powers on the two finite local subgroups gives the stated surviving powers, with separate exact checks in both implementations. □

Remark 6.6 (Imprimitive CM type at the A-vertex)*.* The stabilizer of Φ_A in $G = (\mathbb{Z}/42\mathbb{Z})^\times$ is $\{1, 29\}$. Its fixed field is $\mathbb{Q}(\zeta_7)$, and Φ_A is induced from the three-element CM type on that sextic CM field. Thus the larger divisor algebra at the A-vertex is consistent with the fact that this CM type is imprimitive; in particular the associated sixfold is isogenous to the square of a CM threefold.

7 Integral models, polarizations and maps

We first give coordinates for the common principal component. We then separate marked, polarized and unpolarized equivalence. The ideal and unit calculations are also the explicitly cited arithmetic inputs to Paper II.

Proposition 7.1 (An explicit principal RM component)**.** *Write $O = \mathcal{O}_L$, $u = t - 1$, $d = f'(t)$, and*

$$\beta = -t^5 + t^4 + 6t^3 - 2t^2 - 8t - 2, \qquad a = (\beta u^2)^{-1}, \quad b = u^{-3}, \quad \delta = ab = (\beta u^5)^{-1}.$$

The common self-dual lattice $P = \Lambda_{\mathrm{common}}$ has O-basis

$$T = \begin{pmatrix} a & 3b \\ 0 & b \end{pmatrix}, \qquad P = TO^2. \tag{7.1}$$

Its determinant δ generates the codifferent and has signs $(+,-,+,-,+,-)$ in the ordered embeddings $1,5,11,13,17,19$. A principal RM presentation of the family in Theorem 6.4 has homology module

$$P_{\mathrm{std}} = Oe_1 + \delta Oe_2$$

with the trace form E_1, period lattice $Ow + \delta O$, and arithmetic group

$$\Gamma_{\mathrm{std}} = \left\{ \begin{pmatrix} r & \delta s \\ v/\delta & q \end{pmatrix} : r,s,v,q \in O,\ rq - sv = 1 \right\} = \mathrm{diag}(\delta,1)\,\mathrm{SL}_2(O)\,\mathrm{diag}(\delta,1)^{-1}. \tag{7.2}$$

Its period map on the universal covering curve is

$$F_{\mathrm{std},i}(z) = \frac{\sigma_i(a^2)F_{t,i}(z)}{3F_{t,i}(z)+1}. \tag{7.3}$$

Thus its image lies in the specified principal RM component $\mathbb{H}^6/\Gamma_{\mathrm{std}}$, with a compatible neat refinement when a fine moduli interpretation is required.

Proof. The ideal identities $(u) = \mathfrak{p}_7$, $(\beta) = \mathfrak{p}_3$ and $(d) = \mathfrak{p}_3\mathfrak{p}_7^5$ give the unit

$$\varepsilon = \frac{d}{\beta u^5} = 28 + 215t + 335t^2 + 82t^3 - 87t^4 - 35t^5, \qquad N_{L/\mathbb{Q}}(\varepsilon) = 1.$$

They may also be checked directly from f: $N(u) = 7$, $N(\beta) = -27$ and β vanishes modulo the degree-three factor at 3. Under multiplication by d, the local conditions defining the common subgroup in Proposition 6.1 say

$$y \in \beta O, \qquad y \in u^2 O, \qquad x - 3y \in u^3 O.$$

Since (β) and (u) are coprime, their simultaneous solution is

$$dP = Ou^3(1,0) + O\beta u^2(3,1).$$

Now

$$dT = \varepsilon \begin{pmatrix} u^3 & 3\beta u^2 \\ 0 & \beta u^2 \end{pmatrix}, \qquad T^{-1} = \begin{pmatrix} \beta u^2 & -3\beta u^2 \\ 0 & u^3 \end{pmatrix}.$$

The unit factor leaves the module unchanged, proving (7.1) and $O^2 \subset P$. Also $\delta d = \varepsilon$, so $\delta O = \mathfrak{D}_L^{-1}$ and $[P : O^2] = |N(\delta)|^{-1} = 453789$.

In the basis T, the pairing is $E_\delta((r,s),(r',s')) = \mathrm{Tr}_{L/\mathbb{Q}}(\delta(r's - rs'))$. Exact arithmetic gives

$$\delta = -\frac{11}{21} - \frac{209}{21}t - \frac{55}{3}t^2 - \frac{110}{21}t^3 + \frac{33}{7}t^4 + 2t^5.$$

The integral matrix $(\mathrm{Tr}(\delta t^{i+j}))_{0\le i,j\le 5}$ has determinant -1, so its alternating trace form is unimodular. Disjoint rational isolating intervals for all six roots of f, followed by rational interval evaluation of this polynomial, certify the stated signs. The exact identities and signs are checked in `experiments/principal_normalization_v111/explicit_basis.py` and separately, using rational polynomial arithmetic, in `experiments/principal_normalization_v111/review/independent_basis_check.py`.

At a period $z \in \mathbb{H}^6$, interpreted componentwise, the basis columns evaluate to az and $b(3z+1)$. Division by $b(3z+1)$ gives the period

$$\zeta = \frac{a}{b}\frac{z}{3z+1}$$

on the standard O^2 module with form E_δ. Its imaginary-part signs are those of δ; E_δ is positive on this mixed component. Multiplication of the complex coordinate by δ gives $w = \delta\zeta = a^2 z/(3z+1)$ and lattice $Ow + \delta O$. The corresponding homology identification is

$$C = \operatorname{diag}(1, \delta)T^{-1}, \qquad \det C = 1.$$

It maps P onto P_{std} and preserves E_1 exactly. Thus this is a polarized isomorphism, not just an isogeny comparison. The principal module P_{std} is self-dual because δO is the codifferent. Its transition stabilizer for $g \mapsto g^{-\mathsf{T}}$ is precisely (7.2). The form E_1 is positive on its all-upper-half-plane domain.

The composite period matrix is

$$W = \begin{pmatrix} a^2 & 0 \\ 3 & 1 \end{pmatrix} \in \mathrm{GL}_2^+(L), \qquad \det W = a^2.$$

It gives (7.3); its denominators do not vanish on $\mathbb{H}$. The stabilizer calculation below verifies equivariance on the same connected cover as Theorem 6.4. That theorem's algebraization identifies the resulting moduli map with the original principal family, preserving its degree-453789 isogeny and its three specialized K-actions. □

Proposition 7.2 (Explicit transition stabilizer)**.** *For $g = \left(\begin{smallmatrix} r & s \\ v & q \end{smallmatrix}\right) \in \mathrm{SL}_2(O)$ put $Q(g) = v + 3r - 3q - 9s$. Then*

$$\Gamma_H = \{g \in \mathrm{SL}_2(O) : s \in \beta O,\ Q(g) \in uO\}, \qquad W\Gamma_H W^{-1} = \Gamma_{\mathrm{std}} \cap W\,\mathrm{SL}_2(O)W^{-1}.$$

Proof. The homology basis matrix is T, the period matrix is $S = T^{\mathsf{T}}$, and the transition action is $g^{-\mathsf{T}}$. Direct multiplication gives

$$T^{-1}g^{-\mathsf{T}}T = \begin{pmatrix} q + 3s & -(\beta/u)Q(g) \\ -(u/\beta)s & r - 3s \end{pmatrix}.$$

Its integrality is equivalent to the two displayed congruences, because u and β are coprime. Determinant one then gives a lattice automorphism. On the final periods,

$$WgW^{-1} = \begin{pmatrix} r - 3s & a^2 s \\ a^{-2}Q(g) & q + 3s \end{pmatrix}.$$

The identities $a^2/\delta = u/\beta$ and $\delta a^{-2} = \beta/u$ prove the intersection assertion. Consequently (7.3) is equivariant for $Wj(\Delta')W^{-1} \subset \Gamma_{\mathrm{std}}$. The indices 224 and 28 are those of Proposition 6.3; no new degree after neat refinement is inferred. □

7.1 Marked and unmarked distinctions

Proposition 7.3 (Distinct principal points with fixed real multiplication)**.** *Let B_X, for $X = A, B, C$, be the three specified principal fibers of Theorem 6.4, with their fixed $\mathcal{O}_L$ actions. For $X \neq Y$ one has*

$$\mathrm{Hom}_{\mathcal{O}_L}(B_X, B_Y) \otimes_{\mathbf{Z}} \mathbb{Q} = 0.$$

Consequently their images in the principal RM component $\mathbb{H}^6/\Gamma_{\mathrm{std}}$ are pairwise distinct, also after forgetting auxiliary level while retaining the $\mathcal{O}_L$ marking.

Proof. The actual geometric CM types, expressed using $z = \zeta_{42}$, are

$$\Phi_A = \{1, 5, 17, 19, 29, 31\}, \quad \Phi_B = \{1, 5, 23, 25, 29, 31\}, \quad \Phi_C = \{1, 5, 11, 13, 17, 19\}.$$

Their M actions restrict to the same prescribed L action: the generators for B, C satisfy $Z^2 - tZ + I = 0$, and the certified generator $Z_A = (A_t + tI)/(t^2 - 1)$ of Proposition S.11.7 satisfies the same identity and $Z_A^3 = A_t$. Passing to transposes gives the homology actions without changing the scalar t. Thus $z + z^{-1}$ acts as the fixed scalar t in every case. These are the normalized geometric types, not companion types or representatives re-marked by arbitrary Galois automorphisms. The lattice inclusions of Theorem 6.4 are the identity on rational L^2, and the coordinate normalization in Proposition 7.1 commutes with L at each real place. Hence the same marked rational Hodge structures, with their rational M actions, describe the principal fibers.

For completeness, consider two rank-one M rational Hodge structures with this fixed L restriction. After identifying each rational vector space with M, every rational linear map is uniquely

$$F = \sum_{g \in G} m_g g, \qquad m_g \in M, \qquad G = \mathrm{Gal}(M/\mathbb{Q}).$$

Indeed, after splitting M over $\mathbf{C}$, each summand Mg occupies the matrix positions of its embedding permutation. These supports are disjoint and their dimensions sum to $[M : \mathbb{Q}]^2$. Commutation with every $a \in L$ gives $m_g(g(a) - a) = 0$. Since M is a field, only $g \in \mathrm{Gal}(M/L) = \{1, c\}$ can remain, where c is complex conjugation. If $m_g \neq 0$, every conjugate of m_g is nonzero. Its entries cannot cancel those of another summand. Preserving the Hodge subspaces therefore requires the two types to be equal for $g = 1$, or complementary for $g = c$.

The three displayed types are pairwise unequal, and all contain 1 and omit 41, so no two are complementary. Thus every L-linear Hodge map between distinct fibers is zero. An $\mathcal{O}_L$-linear homomorphism of abelian varieties induces such a map on rational homology, and a homomorphism with zero map on homology is zero. This proves the assertion without a primitivity assumption. An identification in the specified principal RM moduli component would be an $\mathcal{O}_L$-linear polarized isomorphism, which is excluded. The unmarked isogeny relation $19\Phi_B = \Phi_C$ from Proposition 5.18 changes the L marking and gives no contradiction. The next proposition treats the unmarked principal images; neither result identifies Mostaed's named image. □

Proposition 7.4 (Three distinct unmarked principal fibers)**.** *The three specified principally polarized fibers B_A, B_B, B_C of Theorem 6.4 are pairwise nonisomorphic as principally polarized complex abelian varieties, after forgetting both the fixed $\mathcal{O}_L$ action and auxiliary level. Thus they give three distinct points of the coarse moduli space of principally polarized abelian sixfolds.*

Proof. By Proposition 5.18, the underlying A fiber is isogenous to $J(y^2 = x^7 - 1)^2$, whereas the B, C fibers are isogenous to the simple sixfold X_{21}. The principal-model isogenies preserve this distinction, so only B_B versus B_C remains.

Write $z = \zeta_{42}$, $t = z + z^{-1}$, $u = t - 1$ and

$$\beta = -t^5 + t^4 + 6t^3 - 2t^2 - 8t - 2, \qquad a = (\beta u^2)^{-1}, \quad b = u^{-3}.$$

The actual common lattice of Proposition 7.1 is $P = \mathcal{O}_L(a, 0) + \mathcal{O}_L(3b, b)$, with the restriction of E_1. For $X = B, C$, let (c_X, d_X) be the bottom row of the normalized vertex matrix X_t. Explicitly,

$$c_B = t^{-1}, \quad d_B = t, \qquad c_C = t^{-1}(-1 + 5t + 4t^2 - t^3 - t^4), \quad d_C = -t + 4t^2 + 4t^3 - t^4 - t^5.$$

On homology, z acts by X_t^T. The identification $J_X : M \longrightarrow L^2$ given by $J_X(v_0 + v_1 z) = (c_X v_1, v_0 + d_X v_1)$ therefore has

$$I_X = J_X^{-1}(P) = \mathcal{O}_L a w_X + \mathcal{O}_L b(1 + 3w_X), \qquad w_X = (z - d_X)/c_X.$$

In these coordinates its actual polarization is

$$E_X(v, w) = \mathrm{Tr}_{M/\mathbb{Q}}(\xi_X v \bar{w}), \qquad \xi_X = -\frac{c_X}{z - z^{-1}}.$$

Indeed the relative trace equals $c_X(v_0 w_1 - v_1 w_0)$, which recovers $E_1(J_X v, J_X w)$ after applying $\mathrm{Tr}_{L/\mathbb{Q}}$.

The disjoint-support argument in Proposition 7.3 applies without imposing L-linearity: a term $m_g \sigma_g$ can preserve the Hodge subspaces only if $\Phi_B = g\Phi_C$. For the actual types displayed there, the unique such exponent is $g = 31$ modulo 42. Hence every possible isomorphism has the form $F(v) = m\sigma_{31}(v)$. Its integrality and polarization conditions imply, respectively,

$$m \in J := \frac{I_C}{\sigma_{31}(a w_B)}, \qquad m\bar{m} = r := \frac{\sigma_{31}(\xi_B)}{\xi_C}.$$

The first condition uses only the image of the first actual lattice generator; it requires no invertibility of a nonmaximal-order ideal. The second follows by moving σ_{31} through the absolute trace. In particular $q(m) := \mathrm{Tr}_{M/\mathbb{Q}}(m\bar{m}/r) = 12$ is necessary.

We record the finite obstruction explicitly. Let Q be the rational Gram matrix of q on the basis of J induced by $t^j a w_C, t^j b(1 + 3w_C)$, $0 \le j < 6$, divided by $\sigma_{31}(a w_B)$. The certificate `certificates/principal_unmarked_bc_v122.py` reconstructs this matrix from the displayed field and polarization formulas. Its data file `certificates/principal_unmarked_bc_data.json` supplies an integral matrix U with determinant -1 and the exact identity $U^T Q U = H$, where

$$H = \begin{pmatrix}
14 & -7 & 0 & 0 & -6 & -2 & 3 & 2 & 0 & 0 & -3 & 6 \\
-7 & 14 & 0 & 0 & 3 & 4 & -6 & -1 & 0 & 0 & -3 & -3 \\
0 & 0 & 14 & 0 & 4 & -1 & -2 & -6 & 0 & -7 & 2 & -4 \\
0 & 0 & 0 & 14 & -1 & -6 & 2 & -2 & -7 & 0 & 1 & 1 \\
-6 & 3 & 4 & -1 & 16 & 4 & -8 & -8 & 2 & -2 & -1 & 2 \\
-2 & 4 & -1 & -6 & 4 & 16 & -8 & 4 & 3 & 2 & 2 & 2 \\
3 & -6 & -2 & 2 & -8 & -8 & 16 & 4 & -1 & 4 & -1 & -1 \\
2 & -1 & -6 & -2 & -8 & 4 & 4 & 16 & 4 & 3 & -1 & 2 \\
0 & 0 & 0 & -7 & 2 & 3 & -1 & 4 & 14 & 0 & 1 & -2 \\
0 & 0 & -7 & 0 & -2 & 2 & 4 & 3 & 0 & 14 & 2 & 2 \\
-3 & -3 & 2 & 1 & -1 & 2 & -1 & -1 & 1 & 2 & 22 & -11 \\
6 & -3 & -4 & 1 & 2 & 2 & -1 & 2 & -2 & 2 & -11 & 22
\end{pmatrix}.$$

Exact rational factorization proves H positive definite, and direct inversion gives $12(H^{-1})_{ii} < 4$ for every i. Cauchy–Schwarz therefore places every integer vector of norm at most 12 in $\{-1, 0, 1\}^{12}$. Exact enumeration of this box gives only the origin in that closed ball. The primary certificate instead uses rational completion of squares, visiting 59 nodes with the same result. The separate implementation `independent/principal_unmarked_bc_v122.py` reconstructs Q by cyclotomic companion matrices and checks all $3^{12} = 531441$ box vectors by an integer split quadratic form. Both paths verify the unimodular change of basis and the complete bounds. In fact the lattice minimum is 14: H is even integral, the closed ball of norm 12 contains only zero, and its first basis vector has norm 14.

Thus the necessary trace norm 12 is impossible, excluding every polarized isomorphism $B_B \simeq B_C$. This leaves their established unmarked isogeny relation unchanged. The stronger unpolarized distinction is proved in Proposition 7.6 below; neither argument asserts injectivity of the whole principal curve or identifies it with Mostaed's named image. □

7.2 Scalar-polarized maps

Proposition 7.5 (The smallest scalar-polarized B–C isogeny)**.** *For the specified principal fibers (B_B, λ_B) and (B_C, λ_C), the smallest positive integer k for which an isogeny $f : B_B \to B_C$ satisfies $f^*\lambda_C = k\lambda_B$ is 2. There are exactly six such isogenies at $k = 2$, each of degree 64. They have the same kernel, a maximal isotropic subgroup of $B_B[2]$ isomorphic to $(\mathbb{Z}/2\mathbb{Z})^6$, and form one orbit under the polarized automorphisms of B_C. The origin-preserving polarized automorphism groups of both fibers are the explicit cyclic groups of order six generated, in their respective CM coordinates, by multiplication by z^7.*

Proof. Use $I_B, I_C, \xi_B, \xi_C, r, J, Q, H$ and the unique Hodge twist σ_{31} from Proposition 7.4. A rational Hodge map $m\sigma_{31}$ is an isogeny with multiplier k if and only if

$$m\sigma_{31}(I_B) \subseteq I_C, \qquad m\bar{m} = kr.$$

The inclusion must be tested on the full integral lattices; membership in J alone is insufficient. The norm equation is nonzero, so it makes the rational map invertible. An integral Hodge map between these abelian varieties is algebraic, as in the homology comparison above. On the common integral symplectic form E, its matrix satisfies $T^tET = kE$, so Pfaffians give its degree $\det T = k^6$. Moreover, any rational scalar multiplier is integral: unimodularity makes E primitive, so integer linear combinations of its entries equal 1, and the corresponding combinations of the entries of T^tET equal k. Positivity then gives $k > 0$.

For $k = 2$ the necessary trace norm is 24. The exact inverse-diagonal bounds for H give

$$(|n_1|, \ldots, |n_{12}|) \le (1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1).$$

The certificate `certificates/principal_bc_isogeny_v123.py` and the separate companion-matrix implementation `independent/principal_bc_isogeny_v123.py` prove completeness by rational suffix recursion and Schur-complement prefix minima, respectively. The independent path records 175 vectors in the closed trace-norm ball of value 24, and both find 84 on its boundary. Of the latter, 72 fail the full relative norm equation, and six more fail lattice integrality. The remaining six satisfy the full norm equation, all 144 matrix integrality conditions, $T^tET = 2E$ and $\det T = 64$.

Explicitly, one map is $F_0(v) = m_0\sigma_{31}(v)$, where

$$\begin{aligned} 127m_0 = 19 - 98z - 85z^2 - 37z^3 + 15z^4 + 48z^5 - 56z^6 \\ + 24z^7 + 7z^8 + 105z^9 + 82z^{10} + 2z^{11}. \end{aligned}$$

Its integral matrix is retained in `certificates/principal_bc_isogeny_data.json`; the implementations reconstruct it directly from the actual bases of I_B, I_C and verify these identities. The complete six-map list is $F_j = z^{7j}F_0$, $0 \le j < 6$. Multiplication by z^7 preserves each actual integral lattice and its polarization and has exact order six; both implementations check these statements directly. Thus all six maps have the same kernel.

The identity $T^tET = 2E$ gives the integral matrix $2T^{-1} = E^{-1}T^tE$. Hence the cokernel of T is killed by 2; its order 64 makes it $(\mathbb{Z}/2\mathbb{Z})^6$. It identifies with the kernel of the isogeny in $B_B[2]$. For representatives v, w of kernel points, $2E_B(v, w) = E_C(F_0v, F_0w)$ is integral, so their 2-Weil pairing is trivial. Its order 2^6 makes this isotropic subgroup maximal. The independent certificate also computes the Smith form $\mathrm{diag}(1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2)$ for every map.

Postcomposition with any polarized automorphism of B_C permutes the six maps faithfully, since F_0 is surjective. Its orbit of F_0 therefore has the size of that automorphism group and at

most six elements. The six displayed units already occur, proving that they are the full group. The same argument by precomposition applies to B_B: if $F_0 a = F_0 b$, the connected image of the homomorphism $a - b$ lies in the finite kernel of F_0, so $a = b$.

Finally multiplier one would have degree one and give a polarized isomorphism, excluded by Proposition 7.4. Thus multiplier two is minimal. This does not assert a minimum degree among arbitrary unpolarized isogenies, nor a fixed-$\mathcal{O}_L$-linear map; the twist σ_{31} changes the prescribed real multiplication. □

7.3 Unpolarized isomorphisms and minimum degree

Proposition 7.6 (Three distinct underlying abelian varieties)**.** *The three specified principal models B_A, B_B, B_C of Theorem 6.4 are pairwise nonisomorphic as complex abelian varieties, even after forgetting their polarizations and real multiplication markings.*

Proof. The accepted simple-factor decomposition distinguishes B_A from B_B, B_C. For the latter pair, use the actual CM-coordinate lattices I_B, I_C and the rational Hodge twist σ_{31} from Proposition 7.4. Write $S_X = \mathcal{O}_M I_X$ for their maximal-order saturations, and let $F_0 = m_0\sigma_{31}$ be the explicit degree-64 isogeny of Proposition 7.5. Put

$$s = z^6 + z^{12} + z^{24} - z^{18} - z^{30} - z^{36}, \qquad \alpha = \frac{1+s}{2}, \qquad \varpi = 1 - z^{14}.$$

Then $s^2 = -7$, $\alpha\bar{\alpha} = 2$, and exact lattice calculations give

$$F_0(S_B) = \alpha S_C, \qquad [S_X : I_X] = 27, \qquad [\mathcal{O}_M : R] = 27, \quad R = (I_C : I_C), \quad v - \bar{v} \in \varpi\mathcal{O}_M \quad (v \in R). \tag{7.4}$$

The first equality includes an integral determinant-one basis change; an equality of indices alone would not suffice. The last condition is verified on a basis of the *full* multiplier order, not merely a selected suborder.

Here is a finite exact description of the computation. In the basis $1, z, \dots, z^{11}$, the saturation is the lattice generated by $z^i I_X$, $0 \le i < 12$. For each basis element of M, write its multiplication matrix on I_C and collect the 144 entries as a column of a rational matrix A. The full multiplier order consists exactly of the coefficient vectors x for which $Ax \in \mathbb{Z}^{144}$; it is the dual of the lattice generated by the rows of A. The certificate `certificates/principal_bc_unpolarized_v125.py` reconstructs these lattices by integer Hermite forms after clearing denominators. The separate certificate `independent/principal_bc_unpolarized_v125.py` checks the saturations by containment, stability under z and generation modulo 3, and the full multiplier order by linear integrality constraints modulo 3. Both verify (7.4) on the actual C56 lattices and the displayed F_0.

The ideal $\mathfrak{P} = \varpi\mathcal{O}_M$ is the prime above 3 with residue field $\mathbb{F}_{729}$. Indeed $\zeta_3 = z^{14}$ reduces to 1, and 3 has order 6 modulo 7, so Φ_7 is irreducible over $\mathbb{F}_3$. Complex conjugation acts on this field by $x \mapsto x^{27}$, since $27 \equiv -1 \pmod 7$.

Suppose now that $F = m\sigma_{31}$ were an unpolarized isomorphism $B_B \to B_C$. It would send I_B onto I_C and hence S_B onto S_C. Comparing this with (7.4) shows that

$$\eta S_C = S_C, \qquad \eta = \alpha m/m_0.$$

Multiplication by η and by η^{-1} therefore has an integral matrix on the full lattice S_C. Their characteristic polynomials show that both are algebraic integers: $\eta \in \mathcal{O}_M^\times$. This uses no class-number-one or principal-ideal choice.

Let $\mathbb{Z}_{(3)}$ denote localization at the rational prime 3. Since $\det F_0 = 64$, the maps F and F_0 are both isomorphisms from $I_B \otimes \mathbb{Z}_{(3)}$ to $I_C \otimes \mathbb{Z}_{(3)}$. Thus multiplication by $u = m/m_0$ belongs to

$(R \otimes \mathbb{Z}_{(3)})^\times$. By the last condition in (7.4), its nonzero residue is fixed by complex conjugation. Consequently

$$\frac{\eta}{\bar{\eta}} \equiv \frac{\alpha}{\bar{\alpha}} \equiv s \pmod{\mathfrak{P}}.$$

The second congruence uses $s^2 = -1$ in characteristic 3: $(1+s)/(1-s) = s$. The residue of s has exact order 4. On the other hand, the CM unit $\eta/\bar{\eta}$ has absolute value 1 at every complex embedding, so Kronecker's theorem makes it a root of unity in M, hence an element of μ_{42}. In characteristic 3 its reduction has order dividing 14. The two orders are incompatible, proving the contradiction. The degree-64 isogenies remain valid. The next proposition determines the least degree among all unpolarized isogenies. □

Proposition 7.7 (The least unpolarized isogeny degree)**.** *Among all isogenies $B_B \to B_C$ of the specified principal models, with no polarization compatibility imposed, the least degree is 7. There is an explicit such isogeny with cyclic kernel of order 7.*

Proof. Retain I_X, S_X, m_0, α from the preceding propositions and put $\beta = 1 + z - z^3$. Its absolute norm is 7. Define

$$m_7 = z^5 \beta \alpha^{-1} m_0, \qquad F_7 = m_7 \sigma_{31}.$$

Explicitly,

$$\begin{aligned} 127 m_7 = {} & -40 - 41z - 55z^2 - 9z^3 + 62z^4 + 46z^5 + 31z^6 \\ & + 23z^7 + 12z^8 + 53z^9 + 68z^{10} - 51z^{11}. \end{aligned}$$

The matrix $I_C^{-1} M(m_7)\sigma_{31}(I_B)$ is integral with determinant 7. Thus the rational Hodge map F_7 induces an algebraic isogeny of degree 7, with cyclic kernel. Its exact integral matrix is retained in `certificates/principal_bc_min_degree_data.json`. The certificate `certificates/principal_bc_min_degree_v126.py` checks every entry and the determinant; the separate companion implementation `independent/principal_bc_min_degree_v126.py` reconstructs the map independently and checks the Smith form.

For the lower bound, any isogeny $F = m\sigma_{31}$ sends I_B into I_C, and hence sends S_B into S_C. The exact identity $F_0(S_B) = \alpha S_C$ gives

$$F(S_B) = \eta S_C \subseteq S_C, \qquad \eta = \alpha m / m_0.$$

Its integral multiplication matrix makes η an algebraic integer. Since $[S_B : I_B] = [S_C : I_C] = 27$, lattice indices yield

$$\deg F = [I_C : F(I_B)] = [S_C : F(S_B)] = [S_C : \eta S_C] = N_{M/\mathbb{Q}}(\eta).$$

The last norm is positive because M is CM. If it were between 2 and 6, the ideal (η) would have a prime ideal factor of norm 2, 3, 4 or 5. None exists. Indeed $\mathcal{O}_M = \mathbb{Z}[z]$, and exact polynomial arithmetic gives

$$\gcd(\Phi_{42}, X^4 - X) = 1 \quad \text{over } \mathbb{F}_2, \qquad \gcd(\Phi_{42}, X^p - X) = 1 \quad \text{over } \mathbb{F}_p, \quad p = 3, 5.$$

These exclude residue degrees one or two above 2 and degree one above 3 or 5, exhausting the possible prime-power norms at most 6. The primary certificate checks the three gcds; the independent certificate evaluates Φ_{42} on $\mathbb{F}_4$, $\mathbb{F}_3$ and $\mathbb{F}_5$ and verifies that there are no roots. Degree one is excluded by Proposition 7.6. Together with F_7, this proves the asserted minimum.

The map F_7 is not scalar-polarized for the specified principal polarizations: a positive rational scalar multiplier would be an integer k and give degree k^6, which cannot equal 7. This is compatible with the minimum scalar multiplier 2 and degree 64 in Proposition 7.5. We assert no finite census of all unpolarized isogenies. □

7.4 The integral Hom ideal

Proposition 7.8 (Integral maps and an ideal class of order two)**.** *Set* $R = (I_C : I_C) = \mathcal{O}_L + \mathfrak{P}$, *where* $\mathfrak{P} = (1 - z^{14})\mathcal{O}_M$, *and define*

$$\mathfrak{a} = \alpha\mathcal{O}_L + \mathfrak{P}, \qquad \gamma = z^6(1 - z^2).$$

All integral homomorphisms $B_B \to B_C$ *are, uniquely,*

$$F_\eta = \frac{m_0}{\alpha}\eta\sigma_{31}, \qquad \eta \in \mathfrak{a}.$$

The normalized module $\mathfrak{a}$ *is an invertible fractional* R*-ideal under target postcomposition, with*

$$[\mathcal{O}_M : \mathfrak{a}] = 27, \qquad \mathfrak{a}\bar{\mathfrak{a}} = R, \qquad \mathfrak{a}^2 = \gamma R, \qquad [\mathfrak{a}] \in \mathrm{Pic}(R) \text{ of exact order } 2.$$

For every nonzero η, *the map* F_η *is an isogeny of degree* $N_{M/\mathbb{Q}}(\eta)$. *If* $E_{B,q}(v, w) = \mathrm{Tr}_{M/\mathbb{Q}}(\xi_B q v\bar{w})$, *then its actual pullback on the source lattice is*

$$F_\eta^* E_C = E_{B,\,\sigma_{19}(\eta\bar{\eta})}.$$

*In particular the degree-*7 *map of Proposition 7.7 has pullback polarization type* $(1, 1, 1, 1, 1, 7)$.

Proof. Every rational Hodge map has the unique form $m\sigma_{31}$. The map $(m_0/\alpha)\sigma_{31}$ identifies S_B with S_C by Proposition 7.6. Thus an integral map must have $\eta S_C \subseteq S_C$, so $\eta \in \mathcal{O}_M$. Away from 3, both original lattices equal their saturations locally, because their saturation indices are 27; there is no further local restriction there. At 3, the map F_0 is a lattice isomorphism and α is a unit. The remaining condition is exactly

$$\eta/\alpha \in R \otimes \mathbb{Z}_{(3)}.$$

Since $R = \mathcal{O}_L + \mathfrak{P}$ has residue field $\mathbb{F}_{27}$ inside $\mathcal{O}_M/\mathfrak{P} = \mathbb{F}_{729}$, this says that the residue of η lies in the line $\alpha\mathbb{F}_{27}$. Its full inverse image in $\mathcal{O}_M$ is precisely $\mathfrak{a}$. This proves the description of all maps, including sufficiency.

These module equalities also have direct finite certificates. Collecting the 144 entries of the map for each coefficient of η gives rational linear integrality constraints. The primary certificate `certificates/principal_bc_hom_v127.py` takes the exact dual of their row lattice; the result is $\mathfrak{a}$, not just a sublattice of candidate maps. The independent implementation `independent/principal_bc_hom_v127.py` derives the same module from linear conditions modulo 3, with containment and index checks. The primary additionally checks $R = \mathcal{O}_L + \mathfrak{P}$; both verify the full products $\mathfrak{a}\bar{\mathfrak{a}} = R$ and $\mathfrak{a}^2 = \gamma R$ on integral bases. They check that $\gamma \in \mathcal{O}_M^\times$ has absolute norm 1. The first product gives an explicit inverse fractional ideal. If $\mathfrak{a} = \delta R$ were principal, the equality $[\mathcal{O}_M : \mathfrak{a}] = [\mathcal{O}_M : R] = 27$ would give $N_{M/\mathbb{Q}}(\delta) = 1$. Since $\delta \in \mathfrak{a}$, the corresponding map would be an isomorphism, contradicting Proposition 7.6. Hence the class is nontrivial, and the second product proves that its order is exactly two. The full Picard group is determined in Proposition 7.9 below.

The degree formula is the saturation-index argument of Proposition 7.7. For the polarization, the identity $m_0\bar{m}_0 = 2r$ and $\alpha\bar{\alpha} = 2$ gives $m\bar{m} = r\eta\bar{\eta}$. Moving σ_{31} through the trace therefore gives

$$E_C(F_\eta v, F_\eta w) = \mathrm{Tr}_{M/\mathbb{Q}}\big(\xi_B \sigma_{19}(\eta\bar{\eta}) v\bar{w}\big),$$

since $\sigma_{19} = \sigma_{31}^{-1}$. The inverse twist is essential on source coordinates. For nonzero η its multiplier is totally positive; integrality follows from the integral map. The independent checker verifies all 78 quadratic coefficients of this identity on a basis of $\mathfrak{a}$, including the mixed terms. For the degree-7 witness, both paths check the alternating Smith invariants $(1^{10}, 7^2)$, giving the displayed polarization type. The formula describes exactly the pullbacks arising from this Hom module; it does not assert that every totally positive element is realized. □

7.5 Ideal classes and principal polarizations

Proposition 7.9 (The full Picard group of the principal-model order)**.** *For the order $R = \mathcal{O}_L + \mathfrak{P}$ in Proposition 7.8,*

$$\mathrm{Pic}(R) \cong \mathbb{Z}/2\mathbb{Z}, \qquad \mathrm{Pic}(R) = \{[R], [\mathfrak{a}]\}, \qquad \mathfrak{a} = \alpha\mathcal{O}_L + \mathfrak{P}.$$

Here $\mathrm{Pic}(R)$ *denotes invertible fractional R-ideals modulo principal fractional ideals, without an orientation or positivity condition.*

Proof. Write $O = \mathcal{O}_M$. The conductor and residue computations above give

$$O/\mathfrak{P} = \mathbb{F}_{729}, \qquad R/\mathfrak{P} = \mathbb{F}_{27}.$$

Masley's Main Theorem gives class number one for $M = \mathbb{Q}(\zeta_{42}) = \mathbb{Q}(\zeta_{21})$: $\varphi(21) = 12 \le 24$, and 21 is absent from its exceptional table [5, p. 179]. This concerns the maximal order O; the following argument determines the class group of the proper order R.

Let I be an invertible fractional R-ideal. Since IO is principal, rescale I so that $IO = O$. Then

$$\mathfrak{P} \subseteq I \subseteq O: \qquad \mathfrak{P} = \mathfrak{P}(IO) \subseteq I$$

because $\mathfrak{P}O \subseteq R$. Away from the conductor maximal ideal $\mathfrak{P}$ of R, the order is maximal and the normalized ideal equals O locally. At $\mathfrak{P}$, invertibility gives $I_{\mathfrak{P}} = uR_{\mathfrak{P}}$, where normalization makes u a unit of $O_{\mathfrak{P}}$. Therefore $I/\mathfrak{P}$ is a nonzero $\mathbb{F}_{27}$-line in $\mathbb{F}_{729}$. Conversely, the inverse image J_a of any such line $a\mathbb{F}_{27}$ is locally principal: at $\mathfrak{P}$ it is $uR_{\mathfrak{P}}$ for a lift u of a, and elsewhere it is O. Thus every such J_a is invertible, and $J_aJ_b = J_{ab}$ by the same local calculation.

Any isomorphism between fractional ideals is multiplication by an element of $M^\times$. Between normalized ideals it must preserve their O-saturations, so it is multiplication by a global O-unit. Consequently this classifies every invertible ideal class and gives

$$\mathrm{Pic}(R) \cong \frac{\mathbb{F}_{729}^\times}{\mathbb{F}_{27}^\times \,\mathrm{im}(O^\times)}.$$

The intermediate quotient $Q = \mathbb{F}_{729}^\times/\mathbb{F}_{27}^\times$ is cyclic of order 28. Conjugation induces $x \mapsto x^{27}$ on the residue field, and the ratio map

$$Q \longrightarrow \mathbb{F}_{729}^\times, \qquad [x] \longmapsto x/x^{27}$$

is injective, since its kernel consists of the fixed-field units.

For every $\varepsilon \in O^\times$, Kronecker's theorem gives $\varepsilon/\bar{\varepsilon} \in \mu_{42}$. The reduction of μ_{42} has order 14, so the image of the entire global unit group in Q is contained in the subgroup of order 14. It equals that subgroup. Indeed z has residue order 14, and its ratio z^2 has order 7. The global unit $\gamma = z^6(1 - z^2)$ has norm 1 and

$$\gamma/\bar{\gamma} = -z^{14} \equiv -1 \pmod{\mathfrak{P}},$$

so its class has order 2. These two classes generate a subgroup of order 14. The remaining quotient therefore has order 2.

Finally, $\alpha\bar{\alpha} = 2$ makes the residue of α nonzero. With $s = \sqrt{-7}$ chosen as above,

$$\alpha/\bar{\alpha} \equiv s \pmod{\mathfrak{P}}, \qquad s^2 \equiv -1 \pmod{\mathfrak{P}}.$$

Its ratio has order 4, so its class lies outside the global-unit image of order 14. Its residue line has inverse image $\mathfrak{a} = \alpha\mathcal{O}_L + \mathfrak{P}$. This identifies the nonidentity class and agrees with the ideal product identities of Proposition 7.8.

The finite inputs are checked by `certificates/principal_order_class_v129.py`, using Frobenius and polynomial gcds over $\mathbb{F}_3$ and exact rational resultants. The separate implementation `independent/principal_order_class_v129.py` checks all 39 possible monic divisors of degrees at most three, enumerates the 729 residue elements, and computes the global norms by rational multiplication matrices. Both identify the residue field using

$$\Phi_{42}(X) \bmod 3 = (X^6 - X^5 + X^4 - X^3 + X^2 - X + 1)^2.$$

These finite checks supplement the class-number theorem and the ideal and global-unit arguments above; they do not enumerate the infinite group of global units. □

Proposition 7.10 (The ordinary ideal-class monoid of the selected order)**.** *Let $O = \mathcal{O}_M$ and retain R and $\mathfrak{a}$ from Proposition 7.9. Let* $\mathrm{ICM}(R)$ *be the monoid of nonzero fractional R-ideals that are full finitely generated $\mathbb{Z}$-lattices in M, modulo R-linear isomorphism, with ordinary ideal multiplication. Then*

$$\mathrm{ICM}(R) = \{[R], [\mathfrak{a}], [O]\}, \qquad [\mathfrak{a}]^2 = [R], \quad [\mathfrak{a}][O] = [O]^2 = [O].$$

Its unit group is $\mathrm{Pic}(R) \cong \mathbb{Z}/2\mathbb{Z}$*; $[O]$ is the remaining, absorbing idempotent. The multiplier rings $(I : I) = \{x \in M : xI \subseteq I\}$ are R for the first two classes and O for the third. These are ordinary rank-one classes: the zero ideal is excluded, and equivalence is R-linear, without orientation or semilinear identifications. This statement for the selected order makes no geometric realization or polarization claim.*

Proof. Use the conductor $\mathfrak{P}$, residue fields $k = R/\mathfrak{P} = \mathbb{F}_{27}$ and $k_2 = O/\mathfrak{P} = \mathbb{F}_{729}$, and $\mathrm{Pic}(O) = 1$ established in the preceding proposition. For any ideal I in the stated class, IO is a nonzero fractional ideal of the Dedekind domain O. It is principal, even when I is not invertible over R. A scalar rescaling therefore gives $IO = O$ and $I \subseteq O$. Write $1 = \sum_j i_j o_j$ with $i_j \in I$ and $o_j \in O$. For $p \in \mathfrak{P}$,

$$p = \sum_j i_j(po_j) \in I,$$

since $po_j \in \mathfrak{P} \subseteq R$. Thus $\mathfrak{P} \subseteq I \subseteq O$, and $I/\mathfrak{P}$ is a nonzero k-subspace of k_2; otherwise $I = \mathfrak{P}$ would contradict $IO = O$. Since $[k_2 : k] = 2$, this subspace is either a line or all of k_2. The latter case gives $I = O$.

For $t \in k_2^\times$, let J_t be the inverse image of tk. A lift of t, together with $\mathfrak{P}$, generates O as an O-ideal, so $J_tO = O$. The product J_tJ_u also has saturation O and hence contains $\mathfrak{P}$ by the same argument. Its residue is exactly tuk, giving

$$J_tJ_u = J_{tu}, \qquad J_1 = R, \qquad J_tJ_{t^{-1}} = R.$$

Every line lift is therefore invertible over R.

An R-linear isomorphism between these full rank-one ideals extends over $\mathrm{Frac}(R) = M$ to multiplication by $c \in M^\times$. Between normalized ideals it satisfies $cO = O$, so $c \in O^\times$. The preceding proposition gives exactly two line classes, represented by R and $\mathfrak{a}$. The class of O is distinct, since an O-unit cannot carry O to a proper line lift. Normalization gives $[I][O] = [O]$ for every class, and the remaining products follow from $\mathrm{Pic}(R)$.

Finally, $xI \subseteq I$ for a normalized ideal implies $xO \subseteq O$, hence $x \in O$. For $I = J_t$, preservation of the residue line tk is equivalent to $x \bmod \mathfrak{P} \in k$, so $(J_t : J_t) = R$; also $(O : O) = O$. These rings are unchanged by scalar rescaling. As $R \neq O$, the third class is not invertible: an inverse ideal H would give $xR = xIH \subseteq IH = R$ for every $x \in (I : I)$, forcing $(I : I) = R$. □

Proposition 7.11 (Marked principal polarizations for the selected order)**.** *Put $O = \mathcal{O}_M$, and retain R and $\mathfrak{a}$ from Proposition 7.8. Fix a marked CM type Φ of M, obtained from $\Phi_0 = \{1, 5, 11, 13, 17, 19\}$ by conjugating a subset T of its six embeddings. Among full fractional R-ideals, up to M-linear polarized isomorphism, the principal polarizations whose Rosati involution on M is complex conjugation are as follows:*

	$[R]$	$[\mathfrak{a}]$	$[O]$	*total at* Φ
$\lvert T\rvert$ *even*	2	2	0	4
$\lvert T\rvert$ *odd*	0	0	1	1

The origin-preserving marked polarized automorphism groups are μ_6 in the first row and μ_{42} in the second. Thus there are 160 *marked polarized classes over the* 64 *marked CM types. This count makes no identification between different types and no assertion that these objects lie on a specified curve.*

Proof. Write $I^\vee = \{x \in M : \mathrm{Tr}_{M/\mathbb{Q}}(xI) \subseteq \mathbb{Z}\}$. An M-compatible alternating form is $E_\xi(x, y) = \mathrm{Tr}_{M/\mathbb{Q}}(\xi x \bar{y})$, with $\bar{\xi} = -\xi$. Its dual lattice is

$$I^{\#_\xi} = \xi^{-1}(\bar{I})^\vee = \xi^{-1}(R^\vee : \bar{I}).$$

Indeed $x\bar{I} \subseteq R^\vee$ is equivalent to $\mathrm{Tr}(x\bar{I}) \subseteq \mathbb{Z}$, since $\bar{I}R = \bar{I}$. For invertible I, this colon equals $R^\vee \bar{I}^{-1}$. Replacing $R^\vee$ by the maximal-order codifferent here would be incorrect for a nonmaximal order.

Set $d = f'(t)$, retain β from Proposition 7.1, and put

$$\kappa_O = \frac{1}{d(z - \bar{z})}, \qquad \kappa_R = \frac{1}{d\beta(z - \bar{z})}.$$

The identity $\beta O = (1 - z^{14})O$ gives $O = \mathcal{O}_L \oplus \mathcal{O}_L z$ and $R = \mathcal{O}_L \oplus \mathcal{O}_L \beta z$. Monogenic relative trace duality, followed by trace transitivity, therefore gives $O^\vee = \kappa_O O$ and $R^\vee = \kappa_R R$. Since $\mathfrak{a}\bar{\mathfrak{a}} = R$, one also has $(\bar{\mathfrak{a}})^\vee = \kappa_R \mathfrak{a}$. Consequently principal compatible parameters on $R, \mathfrak{a}$ are $\kappa_R u$, and those on O are $\kappa_O u$, with $u \in E_L = \mathcal{O}_L^\times$. For the nonmaximal models this follows because the real units of their multiplier ring R are exactly E_L. The conductor identity is certified by `certificates/cm_polarization_conductor_v135.py`: it verifies $\beta = (1 - z^{14})a_0$ with $a_0 = 1 + z + z^5 + z^6 - z^8 + z^{11}$ an integral unit of norm one.

With the convention $E(x, Jx) > 0$, positivity means $\mathrm{Im}\,\phi(\xi) > 0$ for every $\phi \in \Phi$. At Φ_0, the signs of d are $(+, -, +, -, +, -)$ and $\mathrm{Im}\,\phi(z) > 0$. Thus κ_O has odd negative sign parity. Since $N_{L/\mathbb{Q}}(\beta) = -27$, κ_R has even parity. Conjugating an embedding reverses its imaginary sign. Lemma S.11.2 gives exactly the even-parity subgroup as the real-unit signature image, and $[E_L^+ : E_L^2] = 2$. This proves the asserted existence and nonexistence of positive parameters.

We next compute the relevant unit norms. For $e \in R^\times$, Kronecker's theorem gives $e/\bar{e} \in \mu_{42}$. Its residue modulo $\mathfrak{P}$ is one because $R/\mathfrak{P}$ is the conjugation-fixed field $\mathbb{F}_{27}$. The reduction of μ_{42} has order 14, so its kernel is μ_3. Choose $\zeta \in \mu_6$ with $\zeta^2 = e/\bar{e}$. Then $e/\zeta \in E_L$. Conversely $\mu_6 E_L \subset R^\times$, giving

$$R^\times = \mu_6 E_L, \qquad N_{M/L}(R^\times) = E_L^2.$$

By Theorem S.11.3, $N_{M/L}(O^\times) = E_L^+$. For a fixed positive parameter the polarization classes form a torsor under $E_L^+/N_{M/L}(S^\times)$, where $S = (I : I)$. Proposition 7.10 exhausts the three scalar classes and their multiplier rings, so these quotients prove the table. Polarized automorphisms have norm one and are roots of unity. The roots in R are exactly μ_6, while those in O are μ_{42}. Finally $32 \cdot 4 + 32 \cdot 1 = 160$. □

Proposition 7.12 (All principal polarizations on the specified B/C models)**.** *Let B_B, B_C be the two specified principal models, with their original polarizations λ_B, λ_C. Put $\varepsilon_+ = 4 - t^2$. Their scalar ideal classes are respectively $[\mathfrak{a}]$ and $[R]$. On each underlying variety B_X, exactly two isomorphism classes of principal polarizations occur, represented by λ_X and $\lambda_X \circ \iota(\varepsilon_+)$. After forgetting all CM and RM markings these still give exactly four polarized isomorphism classes over the two underlying varieties. Each has origin-preserving polarized automorphism group μ_6.*

Proof. Use the exact I_B, I_C, ξ_B, ξ_C coordinates of Proposition 7.4, and m_0, α from Proposition 7.8. The scalars

$$c_B = 8 + 11z + 4z^2 + 4z^3 + 8z^4 - 5z^5 - 22z^6 - 24z^7 \\ - 2z^8 + 20z^9 + 18z^{10} + 3z^{11}, \qquad c_C = \alpha\sigma_{31}(c_B)/m_0$$

satisfy $c_B I_B = \mathfrak{a}$ and $c_C I_C = R$. The certificate `certificates/actual_bc_polarizations_v135.py` verifies integral unimodular changes of basis for both equalities, and transports each form to $\xi'_X = \xi_X/(c_X \bar{c}_X) = \kappa_R u_X$ with $u_X \in E_L$. It checks the exact alternating matrices and their pullbacks against the original principal matrices. The complete coefficients and basis changes are retained in `certificates/actual_bc_polarizations_data.json`. These are finite identities using one arithmetic implementation; the following classification is a human argument.

There is also a structural reason for c_C. For $D = (m_0/\alpha)\sigma_{31}(I_B)$, the complete Hom calculation gives $(I_C : D) = \mathfrak{a}$. The proved B identification makes $D = (m_0/\alpha)\sigma_{31}(c_B)^{-1}\bar{\mathfrak{a}}$ invertible; here $\sigma_{31}(\mathfrak{a}) = \bar{\mathfrak{a}}$. Multiplying by D and using $\mathfrak{a}\bar{\mathfrak{a}} = R$ gives the C identification.

For $\gamma = z^6(1 - z^2)$ one has

$$\varepsilon_+ = \gamma\bar{\gamma} = 4 - t^2, \qquad \gamma/\bar{\gamma} = -z^{14} \notin \mu_{21}.$$

Thus ε_+ is a totally positive real unit. If it were v^2 with $v \in E_L$, Kronecker's theorem would give $\gamma/v \in \mu_{42}$, forcing $\gamma/\bar{\gamma} \in \mu_{42}^2 = \mu_{21}$, a contradiction. It represents the nontrivial class of E_L^+/E_L^2. Proposition 7.11 now supplies the two compatible principal polarization classes on each model.

These are all principal polarizations, even without a marking. The actual B and C types in Proposition 7.3 each have trivial stabilizer in $\mathrm{Gal}(M/\mathbb{Q})$. The disjoint permutation supports of $\mathrm{End}_{\mathbb{Q}}(M) = \bigoplus_\sigma M\sigma$ then show $\mathrm{End}^0(B_X) = M$. For any polarization $\lambda = \lambda_X a$, its Rosati involution is $x \mapsto a^{-1}\bar{x}a = \bar{x}$, since M is commutative. Thus every polarization is compatible, and every automorphism lies in the multiplier group $R^\times$. Its norm subgroup is E_L^2, so the two classes remain distinct without markings. Proposition 7.6 excludes all cross identifications between B and C. The automorphism groups follow from Proposition 7.11. No corresponding claim is made for the induced A type. □

8 The unit twist and the principal curve image

Proposition 8.1 (A global unit twist and its coarse involution)**.** *On the actual family $(\mathcal{B}, \iota, \lambda_0)$ of Theorem 6.4, the homomorphism $\lambda_1 = \lambda_0 \circ \iota(\varepsilon_+)$ is another principal polarization over the same base C'. After forgetting auxiliary level, the two polarized families are nonisomorphic over the*

identity of C', and twisting twice returns their polarized-family isomorphism class. On the full coarse component of Proposition 7.1, this operation is the nontrivial involution

$$\tau_+(w)_i = \frac{w_i}{\sigma_i(\varepsilon_+)}, \qquad F_{+,i}(z) = \frac{\sigma_i(a^2/\varepsilon_+)F_{t,i}(z)}{3F_{t,i}(z)+1}.$$

The specified B/C fibers receive their second polarization classes from Proposition 7.12.

Proof. Global RM self-adjointness makes λ_1 symmetric. Its Riemann form multiplies each real eigenspace block of the original positive form by $\sigma_i(\varepsilon_+) > 0$. Since $\varepsilon_+^{-1} \in \mathcal{O}_L$, it is a principal polarization. This is a global algebraic polarization homomorphism; no choice of a global ample line bundle is required. The exact marking comparison in Proposition 7.3 identifies $z + z^{-1}$ with the fixed global t, so the B/C twists are literal instances of the preceding proposition.

In the standard module P_{std} put $D = \mathrm{diag}(\varepsilon_+^{-1}, 1)$. It preserves the lattice and satisfies $E_{\varepsilon_+}(Dx, Dy) = E_1(x, y)$. Evaluation at w after D is evaluation at w/ε_+. This gives the displayed period formula with its stated direction. Conjugating $\left(\begin{smallmatrix} r & \delta s \\ v/\delta & h \end{smallmatrix}\right)$ by D gives $\left(\begin{smallmatrix} r & \delta s/\varepsilon_+ \\ \varepsilon_+ v/\delta & h \end{smallmatrix}\right)$, so D normalizes Γ_{std}. Moreover $D^2 = \varepsilon_+^{-1}\,\mathrm{diag}(\varepsilon_+^{-1}, \varepsilon_+)$; the second factor belongs to Γ_{std} and the scalar acts trivially on the domain. This proves the coarse involution. Equivalently, twisting twice is pullback by $\iota(\varepsilon_+)$. The distinct B/C polarization classes witness nontriviality of both the coarse action and the two families over the identity base. □

Proposition 8.2 (The unit involution preserves the principal curve)**.** *The involution τ_+ preserves the entire image V_{std} of (7.3), and its restriction to that image is nontrivial. More precisely, let $a_\triangle, b_\triangle, c_\triangle$ be the original triangle generators, so that $A = a_\triangle^2$, $B = b_\triangle$, $C = c_\triangle^2$, and put $g = BACc_\triangle$. For the actual coarse moduli map $f_0(z) = [WF_t(z)]$ one has*

$$\tau_+ f_0(z) = f_0(g^{-1}z).$$

Proof. The full-triangle equivariance of the actual constructed map is essential. McMullen's Section 4 identifies the quadratic trace-field extension and its nontrivial automorphism, which sends the triangle generators to $(-a_\triangle, b_\triangle, -c_\triangle)$. Thus the two extensions of each real embedding of L give identical projective matrices and ordered target vertices. The vertex-respecting triangle maps in [6, Thm. 6.3, Eq. (6.1)] therefore agree for the two lifts. Keeping one from each pair yields the six components of the actual F, with equivariance for the full triangle group. Conjugating target coordinates by h_t preserves this equivariance. This argument would not follow from arbitrary Δ_0-equivariance.

Set

$$s_0 = -2 - 3t + 4t^2 + t^3 - t^4, \quad h = s_0(I + C_t), \quad k = B_t A_t C_t h.$$

Exact field arithmetic gives $s_0^2 = 2 - t$ and integral inverses for s_0 and $\varepsilon_+ = 4 - t^2$. Consequently $\det h = \det k = \varepsilon_+$ and $h, k \in \mathrm{GL}_2^+(\mathcal{O}_L)$. If $\theta = 2\cos(\pi/42)$, Cayley–Hamilton gives $I + c_\triangle^2 = \theta c_\triangle$. Therefore h is projectively the normalized $c_\triangle$, and $F_t(gz) = kF_t(z)$, componentwise. The first-component base matrix is $h_{t,1}^{-1}k_1h_{t,1}$ projectively; it is not k_1 in the original base marking.

For $D = \mathrm{diag}(\varepsilon_+^{-1}, 1)$ the explicit witness satisfies

$$T^{-1}k^{-\mathsf{T}}T \in \mathrm{GL}_2(\mathcal{O}_L), \qquad U = \mathrm{diag}(\delta, 1)^{-1}DWkW^{-1}\,\mathrm{diag}(\delta, 1) \in \mathrm{SL}_2(\mathcal{O}_L).$$

The exact coefficient arrays, integral inverses and determinant identities are checked by `certificates/curve_twist_image_v138.py` with `certificates/curve_twist_image_data.json`. Only

this fixed word is needed; no search-completeness assertion enters. In particular $\gamma = DWkW^{-1}$ belongs to the specified Γ_{std}. Hence

$$[DWF_t(z)] = [\gamma W k^{-1} F_t(z)] = [WF_t(g^{-1}z)].$$

Since g^{-1} maps $\mathbb{H}$ bijectively to itself, this proves equality of the entire images. They are the actual algebraic images by Theorem 6.4 and Proposition 8.1. The moved B/C points in Proposition 7.12 show that the restricted involution is nontrivial. □

Proposition 8.3 (Two principal classes over each specified RM variety)**.** *For each $X = A, B, C$, the underlying marked variety (B_X, ι_L) has exactly two RM-compatible principal-polarization isomorphism classes, represented by λ_X and $\lambda_X \circ \iota_L(\varepsilon_+)$. Their origin-preserving RM-polarized automorphism groups are μ_6. The resulting six marked objects are pairwise distinct and all lie on V_{std}.*

Proof. Identify rational homology with the rank-one module over the chosen M. This remains valid for the induced A type. Since M/L is quadratic, every L-linear rational endomorphism has the unique form $m_a + m_b\,\overline{(\cdot)}$, for $a, b \in M$. After complexification the first summand preserves each embedding line, while the second exchanges every line with its conjugate. Each CM type contains exactly one line from every conjugate pair; if $b \neq 0$, all its embedding values are nonzero. Thus the second summand cannot preserve the Hodge decomposition or cancel with the first. It follows that

$$\mathrm{End}_L^0(B_X) = M, \qquad \mathrm{End}_L(B_X) = R,$$

where the integral equality uses Proposition 6.5 for all three selected field actions. The first equality is a centralizer statement, not an assertion that the full endomorphism algebra at A is M.

If λ is another RM-compatible principal polarization, then $a = \lambda_X^{-1}\lambda$ is an integral automorphism commuting with L. Symmetry makes a fixed by the original Rosati, hence $a \in E_L$, and positivity gives $a \in E_L^+$. Conversely each such unit supplies a compatible principal polarization. The L-linear automorphisms are precisely $R^\times$, whose norms are E_L^2. Therefore the classes are $E_L^+/E_L^2 = \{1, \varepsilon_+\}$ by Propositions 7.11 and 7.12. The same norm computation gives the stated automorphism groups. Proposition 7.6 excludes identifications between different X, while Proposition 8.2 realizes each second class on the same image at $g^{-1}z_X$. □

Proposition 8.4 (Unmarked separation at the induced A fiber)**.** *The principally polarized varieties (B_A, λ_A) and $(B_A, \lambda_A \circ \iota_L(\varepsilon_+))$ are nonisomorphic after forgetting all CM and RM markings.*

Proof. Use the actual selected generator $Z_A = (A_t + tI)/(t^2 - 1)$. Its lower-left and lower-right entries are

$$c = \frac{t^3 - 1}{t(t^2 - 1)}, \qquad d_0 = \frac{-t^5 + 5t^3 + t^2 - 5t}{t^2 - 1}.$$

In $M = \mathbb{Q}(z)$, set $w = (z - d_0)/c$. The row $(w, 1)$ is an eigenvector of Z_A^{T}, so $(x, y) \mapsto xw + y$ identifies the actual rational homology with M. On the common principal basis of Proposition 7.1, its lattice and polarization parameter are

$$I_A = \sum_{j=0}^{5} \mathbf{Z}\, awt^j + \sum_{j=0}^{5} \mathbf{Z}\, b(1 + 3w)t^j, \qquad \xi_A = -\frac{c}{z - \bar{z}}.$$

The trace form $\mathrm{Tr}_{M/\mathbb{Q}}(\xi_A x\bar{y})$ reproduces the actual principal alternating matrix E exactly.

In $\mathrm{End}_{\mathbb{Q}}(M) = \bigoplus_\sigma M\sigma$, the summands have disjoint permutation supports on the complex embedding lines. A nonzero summand is a Hodge endomorphism precisely when its automorphism stabilizes the actual type $\{1, 5, 17, 19, 29, 31\}$. Its stabilizer is $\{1, 29\}$, and therefore

$$\mathrm{End}^0(B_A) = M + M\sigma_{29}.$$

In particular the full algebra, of rational dimension 24, is used; restricting to multiplication or to one semilinear summand would not prove the assertion.

Let $U_1, \dots, U_{24}$ be the rational homology matrices for z^j and $z^j\sigma_{29}$, $0 \le j < 12$, in the displayed basis of I_A. Form the 144×24 matrix C by flattening them. The full integral endomorphism coefficient lattice is

$$\mathcal{L} = \{v \in \mathbb{Q}^{24} : Cv \in \mathbf{Z}^{144}\}.$$

Equivalently, it is the rational dual of the integer lattice generated by the constraint rows. This gives a complete lattice calculation, including cancellations between the two summands.

Let Q_+ be the homology action of ε_+. An unmarked polarized isomorphism U would satisfy $U^{\mathsf{T}}EU = EQ_+$, hence

$$\mathrm{Tr}_{\mathbb{Q}}(Q_+^{-1}U^\dagger U) = 12, \qquad U^\dagger = E^{-1}U^{\mathsf{T}}E.$$

This is a positive definite quadratic form on the real Hodge endomorphism algebra: the positive Hodge metric makes $\dagger$ the adjoint, and Q_+ is positive self-adjoint. Its exact Gram matrix on $\mathcal{L}$ has an integral unimodular reduction H with positive rational LDL data (μ, d). Every vector in its closed norm-12 ball satisfies $|x_i|^2 \le 12(H^{-1})_{ii}$. The exact decomposition

$$x^{\mathsf{T}}Hx = \sum_i d_i \left(x_i + \sum_{j>i} \mu_{ji}x_j \right)^2$$

therefore supplies finite coordinate bounds and exhaustive suffix pruning using only rational comparisons.

The certificate `certificates/a_unmarked_polarization_v140.py` reconstructs the actual model, full dual lattice and Gram matrix, checks the integral unimodular reduction, positive LDL data and bounds in `certificates/a_unmarked_polarization_data.json`, and exhausts this ball in 347 nodes: only zero occurs. The separate implementation `independent/a_unmarked_polarization_v140.py` works directly on L^2 homology, constructs σ_{29} by a cyclic vector, proves fullness by integer row saturation, and recomputes the Gram and LDL in reverse coordinate order. Its 925 nodes again leave only zero. Thus the required norm-12 vector does not exist. □

Proposition 8.5 (The four B/C polarization pairs)**.** *Put $B_e = (B_B, \lambda_B\varepsilon_+^e)$ and $C_f = (B_C, \lambda_C\varepsilon_+^f)$, for $e, f \in \{0, 1\}$. For every pair $B_e \to C_f$ the least positive integer scalar polarization multiplier is 2, attained by exactly six isogenies, all of degree 64. With $\gamma_0 = z^6(1 - z^2)$ and $v_{ef} = \sigma_{31}(\gamma_0)^e\gamma_0^{-f}$, their normalized Hom parameters are*

$$\eta = \begin{cases} \alpha v_{ef} z^{7j}, & e = f, \\ \bar{\alpha} v_{ef} z^{7j}, & e \ne f, \end{cases} \qquad 0 \le j < 6.$$

The six maps share one kernel for each pair. The pairs $(0,0), (1,1)$ have the same kernel K_+, while $(0,1), (1,0)$ have a second kernel K_-. These are complementary order-64 subgroups of $B_B[2]$.

Proof. Proposition 7.8 gives every rational homomorphism as $F_\eta = (m_0/\alpha)\eta\sigma_{31}$, integral exactly when $\eta \in \mathfrak{a}$. Its source pullback multiplier is $\sigma_{19}(\eta\bar{\eta})$. For the two twisted forms the required multiplier-n equation is consequently

$$\eta\bar{\eta} = n\frac{\sigma_{31}(\varepsilon_+)^e}{\varepsilon_+^f}.$$

Since $\gamma_0\bar{\gamma}_0 = \varepsilon_+$, $N_{M/L}(v_{ef})$ is the unit factor on the right.

The polynomial Φ_{42} modulo 2 is the product of the two irreducible sextics

$$(X^6 + X^4 + X^2 + X + 1)(X^6 + X^5 + X^4 + X^2 + 1).$$

Together with $\alpha\bar{\alpha} = 2$ and $\alpha + \bar{\alpha} = 1$, this identifies the two conjugate primes above 2 as $(\alpha), (\bar{\alpha})$. Every integral solution for $n = 2$ therefore has one of these two ideals. Division by αv_{ef} or $\bar{\alpha}v_{ef}$ gives an integral unit of relative norm one, hence a root of unity. Thus the complete candidates are the two families $\alpha v_{ef} z^j$, $\bar{\alpha}v_{ef}z^j$, $0 \le j < 42$.

For membership in $\mathfrak{a} = \alpha\mathcal{O}_L + \mathfrak{P}$, consider the quotient $\mathbf{F}_{729}^\times/\mathbf{F}_{27}^\times$ of order 28. The class of z has order 7. The class of γ_0 is the unique element of order two: $\gamma_0/\bar{\gamma}_0 = -z^{14}$ reduces to -1. The class of $\bar{\alpha}/\alpha$ is that same element, since $\alpha^2 - \alpha + 2 = 0$ modulo 3 gives $\alpha^4 = -1$ and $(\alpha/\bar{\alpha})^2 = -1$ there. The automorphism σ_{31} fixes this unique order-two class. Consequently the displayed branch cancels the order-two class exactly as stated, and the remaining exponent must be divisible by 7. There are exactly six maps. Their full norm, ideal membership, integral homology matrices and polarization identities are verified in `certificates/bc_polarization_square_v141.py` with its retained data.

Multiplier one would imply an isomorphism of the underlying B/C varieties, excluded by Proposition 7.6. Thus the displayed multiplier two is minimal. Multiplication by $z^{7j} \in R^\times$ preserves each target form, so the six maps for a fixed pair have the same kernel.

Let T_{ef} denote the integral homology matrix for $j = 0$. The certificate verifies $\det T_{ef} = 64$ and the four actual polarization equations. The polarized adjoint relation implies that every kernel lies in $B_B[2]$. It is therefore identified with $\ker(T_{ef} \bmod 2)$ on the common source basis. All four ranks are 6; the stacked ranks for $(00, 11)$ and $(01, 10)$ are 6, and the stacked rank for $(00, 01)$ is 12. These exact finite calculations prove the asserted equality and complementarity of kernels. □

Remark 8.6 (Image and level boundaries). The two parametrized moduli maps differ at the specified B/C parameters, but their images agree by Proposition 8.2. The diagonal matrix in the square calculation need not lie in a chosen neat subgroup, so no involution preserving an arbitrary fine-level marking is asserted. These constructions neither add a point to the original trace-polarized packet nor identify a principal image with Mostaed's literal normalization.

Corollary 8.7 (Six Weil points on the specified principal image)**.** *Let $\mathcal{W}_{K,\mathrm{std}}$ denote the intrinsic K-Weil locus in the specified principal RM component. Then $V_{\mathrm{std}} \cap \mathcal{W}_{K,\mathrm{std}}$ contains the six distinct marked points of Proposition 8.3. All have K-signature $(3, 3)$ and Hermitian discriminant -1. The actual CM types and Hodge algebras are those of their respective A/B/C underlying varieties. All six remain distinct as principally polarized varieties after forgetting the RM and CM markings.*

Proof. Keep the original lattice P, chosen field actions and integral $\mathcal{O}_K$ generators of Proposition 6.1. Twisting replaces E_1 by $E_{\varepsilon_+}(v, w) = E_1(\varepsilon_+ v, w)$, where $\varepsilon_+ I_2$ is an integral automorphism of P. The chosen M action and therefore the actual CM type are unchanged. When the new point is represented at $g^{-1}z_X$, these are the transported actions through the explicit orbit identification above; they are not inferred from a companion CM type.

The integral K action commutes with ε_+ and its Rosati remains conjugation. The complex K action is unchanged, so the signature is still $(3,3)$. On the six-dimensional K homology space, $M = LK$ and $L \cap K = \mathbb{Q}$ give

$$\det_K(m_{\varepsilon_+}) = N_{M/K}(\varepsilon_+) = N_{L/\mathbb{Q}}(\varepsilon_+) = 1.$$

The last norm is also checked in the exact transport certificate. Thus twisting the Hermitian form in one variable leaves its determinant and discriminant unchanged. Positivity and principality were established in Proposition 8.1. Changing polarization does not change the Hodge structure, so the previous divisor/Weil generator and algebraicity conclusions apply. Finally, the four B/C polarized objects remain distinct without markings by Proposition 7.12, and Proposition 8.4 separates the two A objects. The underlying A variety is different from both B/C varieties by Proposition 7.6. Hence all six remain distinct without markings; no classification of all A polarizations is needed. □

Remark 8.8 (The integral comparison still required). For any full O lattice $P' \subset L^2$ self-dual for E_1, its determinant fractional ideal is $\bigwedge_O^2 P' = \mathfrak{D}_L^{-1}$. Indeed at every finite prime a local basis matrix $T_\mathfrak{p}$ turns the pairing into the standard alternating form multiplied by $\det T_\mathfrak{p}$; perfection of the trace pairing is equivalent to $(\det T_\mathfrak{p})O_\mathfrak{p} = \mathfrak{D}_{L,\mathfrak{p}}^{-1}$. These equalities determine the fractional ideal globally. Proposition 4.2 shows that the codifferent has no totally positive generator. Thus no $T' \in \mathrm{GL}_2^+(L)$ satisfies $T'O^2 = P'$. The positive matrix W above compares P to the different module P_{std} and does not contradict this obstruction.

The original trace-polarized fibers do not become principal by passing to a cover alone. The chosen quotient kernel has orbit size 28, so its marked descent requires its stabilizer; this does not rule out descent of an unmarked target or a suitable unspecified finite-index subgroup. In particular Corollary 8.7 concerns the explicit principal image just constructed. Identifying it with Mostaed's named V requires an integral polarization/level comparison and equality of curve images or direct point membership. Neither that identification nor disjointness of the two images follows here. The original standard lattice, a principal isogenous model, and literal membership in a prescribed principally polarized locus remain distinct assertions.

9 Scope and questions for further research

Under the execution premise, the fixed marked packet meets the normalized curve at precisely three quotient points. The common lattice construction and the unit twist give the stated principal models and six Weil points. These conclusions keep the integral lattice, polarization and field marking visible; they do not identify Mostaed's literal principal locus.

There are two natural continuations. First, determine whether the same finite-domain method handles other norm generators and levels. Such an extension needs a new arithmetic domain and new geometric certificates. Second, identify the constructed principal family with a specified moduli normalization, or prove that the integral data obstruct that identification. The latter question must fix the lattice, level and polarization module.

The finite quotient family of Paper II starts from a primitive Fermat factor, not from further points of this packet. Its Jacobian and descent questions therefore do not constitute unfinished cases of the three-point intersection theorem. The supporting Hodge note likewise concerns abstract fixed-field types and does not add points to this curve.

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